Supplemental Material for: Overlapping communities as connected components of a commonality hypergraph

Dmitri Deinega,[1] Igor Kanovsky,[2] and Baruch Barzel[1]

[1] *Department of Mathematics, Bar-Ilan University, Ramat Gan 5290002, Israel*
[2] *Max Stern Yezreel Valley College, Emek Yezreel, Israel*

This Supplemental Material gives the data and benchmark settings, the details of the commonality analysis of Sec. III, and the full tables behind Sec. VII. Every further number (the benchmarks scored by all-community recovery, the selector and grid-edge diagnostics, the SNAP graphs on their induced subgraphs and against distinct node sets, and every method's Ω and coverage on the smaller real networks) is in the repository named in the main text, with the script that produces it.

S1. DATA

Table S1 lists the four networks with recorded memberships, and Table S2 the fourteen Netzschleuder networks of Table S20. Groups of fewer than three nodes are discarded; the SNAP top-5000 lists repeat some node sets, and Ω is scored against the distinct sets.

TABLE S1. Networks with recorded memberships. Groups: the lines of the SNAP files `all.cmty` (Amazon: `all.dedup.cmty`), as used in Sec. III; detection is scored against the top-5000 lists, which repeat some node sets (1 517 distinct sets for Amazon, 4 961 for DBLP); Ω is scored against the distinct sets. On YouTube detection is scored against every group of at least three nodes (8 385 groups on 47 135 nodes). f_{lab}: fraction of nodes with at least one recorded membership.

Network	Nodes	Edges	Groups	f_{lab}
Karate club	34	78	2	1.00
DBLP	317 080	1 049 866	13 477	0.82
Amazon	334 863	925 872	75 149	0.95
YouTube	1 134 890	2 987 624	16 386	0.046

TABLE S2. The fourteen networks of Table S20: nodes, edges, recorded groups, and the fraction of nodes that carry at least one group; groups of fewer than three nodes are discarded. Political blogs includes 266 isolated nodes; the primary school network aggregates both days of contacts.

Network	Nodes	Edges	Groups	f_{lab}
College football	115	613	12	1.00
Political books	105	441	3	1.00
Political blogs	1490	16715	2	1.00
Primary school	242	8317	10	0.96
Facebook ego 0	333	2519	15	0.81
Facebook ego 107	1034	26749	9	0.46
Facebook ego 348	224	3192	14	0.97
Facebook ego 414	150	1693	7	0.89
Facebook ego 686	168	1656	14	1.00
Facebook ego 698	61	270	8	0.84
Facebook ego 1684	786	14024	15	0.97
Facebook ego 1912	747	30025	39	0.95
Facebook ego 3437	534	4813	18	0.18
Facebook ego 3980	52	146	5	0.77

S2. BENCHMARK DESIGN, GRIDS AND SETTINGS

Table S3 lists the ring families, Table S4 the parameter grids and Table S5 the shared nodes and fused rings of the ring benchmarks.

Ring generator. Rings are generated independently. The two nodes of each merged pair are drawn uniformly among all nodes and redrawn when they lie on the same ring or either has already been merged; the absorbed node's edges are moved to the survivor, so the graph has $n_0 - \lfloor o_n n_0 \rfloor$ nodes, $\lfloor o_n n_0 \rfloor$ of them in two rings. A rewired edge keeps one endpoint, chosen at random, and moves the other to a non-neighbour w chosen either uniformly or preferentially, with probability proportional to $(d_w + 1)^\alpha$ under the current degrees ($\alpha = 1$, linear; $\alpha = 2$, superlinear, which grows hubs); as $p \to 1$ the planted structure is erased. The geometry families rewire inside each ring before the merges and the noise, moving one endpoint of each chosen edge to a uniformly random non-neighbour in the same ring; a control family rewires by double-edge swaps inside each ring instead, which keeps every node's degree. In the hub families each node draws its own even degree k_i from a power law $P(k) \propto k^{-2.5}$ on $\{k_r, \ldots, 8k_r\}$ and is joined to its $k_i/2$ nearest neighbours on either side, so that communities keep their lattice neighbourhoods but contain hubs. In the split-overlap families each of the two merged nodes first drops a random half of its edges. Each random geometric community is a random geometric graph on the unit torus with the size and mean degree of a ring, clustered (about 0.58) but not a lattice.

LFR graphs. The reference implementation [1] (`unweighted_undirected`, compiled unmodified) with average degree $\langle k \rangle = 20$, maximum degree $k_{\max} = 50$, degree and community-size exponents $\tau_1 = 2$ and $\tau_2 = 1$, and community sizes between 20 and 100. The three slices at $o_n = 0.1$ use $\langle k \rangle = 10$, $o_m = 4$, and community sizes between 50 and 250, respectively; the two clustered families, also at $o_n = 0.1$, use the generator's clustering option `-C` ($C_{\mathrm{LFR}} = 0.3$, 0.5).

Methods, software and settings. Implementations are those of CDLIB [2] (version 0.4.0), LEIDENALG [3] (0.11.0, with python-igraph 1.0.0), INFOMAP [4] (2.10.1) and GRAPH-TOOL [5] (3.9), except SCAN and link communities, which we implemented to expose the similarity and the cut, and OSLOM, the authors' OSLOM2 code (`oslom_undir`, unweighted); the stochastic block model and OSLOM were run on the same graphs, checked edge for edge. We also use networkx 3.6.1, numpy 2.4.1 and scipy 1.17.1. Leiden: iterated to convergence (`n_iterations`$= -1$), three restarts per resolution on the benchmarks and the fourteen smaller networks, the restart kept being the one of highest objective value under every protocol, so that no label enters the choice of restart, and one run per resolution on Amazon, DBLP and YouTube; under the native protocol the L-CPM resolution equals the edge density, at which the Constant Potts Model compares a group with an Erdős–Rényi graph of the same density [6]. Infomap: two-level, one trial. Link communities: single linkage on closed-neighbourhood Jaccard, cut at every value of HyperCommon's threshold grid, communities of fewer than three nodes discarded; under the native protocol, cut where the partition density is largest over the same grid, the criterion of Ref. [7]. SCAN: cosine similarity on closed neighbourhoods (SCAN-J: the same algorithm with Jaccard similarity); border nodes reachable from several clusters are kept in all of them. Walktrap: walks of length 4, dendrogram cut at maximum modularity (CDLIB's default). Stochastic block model: run under Ubuntu (WSL), `minimize_nested_blockmodel_dl` with default settings, seeded from the graph seed (environment in `paper/environment-sbm.yml`). Stochastic methods are seeded from the graph seed on the benchmarks, use five seeds on the fourteen smaller networks (means reported) and one on Amazon, DBLP and YouTube.

TABLE S3. Ring-lattice families: rings of size s and degree k, written $(s, k)\times$count; $n_0 = 2000$ nodes before the merges, $o_n = 0.05$ unless stated. All six unequal families were also run at $o_n = 0$ and 0.10. Geometry families: uniform $(50, 8) \times 40$, two worlds, three big, many small, and the gradient, each with a fraction $q \in \{0, 0.1, 0.3, 0.6, 1\}$ of every ring's edges rewired inside the ring (one endpoint moved), noise $p \in \{0, 0.1, \ldots, 0.6\}$; the degree-preserving control uses double-edge swaps on uniform $(50, 8) \times 40$ and three big, many small at $q \in \{0.1, 0.3\}$. Isolated-merge families: uniform $(50, 8) \times 40$ and $(100, 16) \times 20$ at $o_n = 0.01$, merges more than $2k$ apart on every ring, $p \in \{0, 0.05, 0.1, 0.2\}$. Hubs: $(50, k_i) \times 40$ with $k_i \in \{4, \ldots, 32\}$, $(100, k_i) \times 20$ with $k_i \in \{8, \ldots, 64\}$, and two worlds with $k_i \in \{16, \ldots, 128\}$ on the five large rings and $\{4, \ldots, 32\}$ on the twenty small ones, $P(k_i) \propto k_i^{-2.5}$ (mean degrees 9.2, 20.6 and 26.5). Split overlap: two worlds and three big, many small at $o_n \in \{0.05, 0.10\}$. Random geometric communities: the sizes and mean degrees of uniform $(50, 8) \times 40$ and of two worlds. These control families use $p \in \{0, 0.1, \ldots, 0.8\}$. Null graphs: random geometric (2000 nodes, mean degree 8) and Holme–Kim (2000 nodes, $m = 4$, triangle probability 0.9), $p = 0$.

Family	Rings
uniform	$(50, 8) \times 40$ or $(100, 16) \times 20$; $o_n \in \{0, 0.05, 0.10\}$
two worlds	$(200, 32) \times 5$, $(50, 8) \times 20$
three big, many small	$(400, 32) \times 3$, $(50, 8) \times 16$
gradient	$(600, 16)$, $(400, 12)$, $(300, 12)$, $(200, 8)$, $(150, 8)$, $(120, 8)$, $(100, 8)$, $(60, 8)$, $(40, 8)$, $(30, 8)$
equal k, unequal s	$(100, 16) \times 10$, $(50, 16) \times 20$
	$(100, 8) \times 10$, $(50, 8) \times 20$
equal s, unequal k	$(100, k) \times 5$, $k \in \{32, 24, 16, 8\}$
preferential	uniform $(50, 8) \times 40$ and two worlds, $\alpha \in \{1, 2\}$

TABLE S4. Parameter grids (Sec. VI; the grids of clique percolation, SLPA, DEMON, ANGEL and SCAN are the widened ones). Logarithmically spaced grids list their end points and number of values. On Amazon, DBLP and YouTube, Leiden's RB resolution runs over $\{0.01, 0.02, 0.05, 0.1, 0.25, 0.5, 1, 2, 4, 8, \ldots, 4096\}$ (19 values, doubling from 1), its Constant Potts Model resolution over 20 values from 10^{-5} to 0.5, Infomap over the same Markov times and the HyperCommon variants over the same t grid; on Amazon and DBLP, link communities also use the t grid, and each commonality function 50 quantile levels of its own triple scores. On the fourteen smaller networks the HyperCommon variants and link communities search 100 quantile levels and the other methods the grids below; on the benchmarks each commonality function searches 29 levels ($0, 0.04, \ldots, 0.96$ and $0.97, 0.98, 0.99, 0.995$).

Method	Grid
HyperCommon, ablations, link communities	$t \in \{0.02, 0.04, \ldots, 0.60\}$ (30)
Leiden, RB modularity	γ from 1/16 to 32 (30, logarithmic), best of 3 restarts
Leiden, Constant Potts Model	γ from 0.002 to 0.5 (30, logarithmic), best of 3 restarts
Infomap	Markov time from 0.25 to 16 (13, logarithmic)
CPM	clique size $\in \{3, \ldots, 7\}$
SLPA	$r \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.3, 0.45\}$, 100 iterations
DEMON	$\epsilon \in \{0.1, 0.25, 0.5, 0.75, 0.9, 1\}$
ANGEL	threshold $\in \{0.1, 0.25, 0.5, 0.75, 0.9, 1\}$
SCAN (cosine)	$\epsilon \in \{0.1, \ldots, 0.8\}$, $\mu_{\mathrm{SCAN}} \in \{2, 3, 5\}$
SCAN (Jaccard)	$\epsilon \in \{0.1, \ldots, 0.6\}$, $\mu_{\mathrm{SCAN}} \in \{2, 3, 5\}$
Walktrap	walks of length 4, dendrogram cut at maximum modularity
label propagation	parameter-free
stochastic block model	level of the nested hierarchy (finest first; 1–6 levels with more than one group)
OSLOM	$t \in \{10^{-14}, 10^{-11}, 10^{-8}, 10^{-5}, 10^{-2}, 0.1\}$, otherwise its defaults (10 runs, first level)

TABLE S5. Shared pairs in the ring benchmarks without noise (mean over ten graphs): merges, ring pairs sharing at least one node, ring pairs sharing two merged nodes within ring distance k on both rings (Proposition 3), the share of rings in such a pair, and the share of rings that exact HyperCommon fuses with another ring just above the lower edge $1/(2k_{\min}+1)$ of the recovery window (a detected community holding at least three unshared nodes of each of two rings; analysis/double_bonds.py).

Family	o_n	merges	sharing	pair	rings	fused
uniform $(50, 8)$	0.05	100	94.2	0.8	0.04	0.11
uniform $(50, 8)$	0.10	200	180.0	2.2	0.11	0.38
uniform $(100, 16)$	0.05	100	77.2	2.6	0.24	0.61
uniform $(100, 16)$	0.10	200	126.9	8.7	0.61	1.00
two worlds	0.05	100	74.7	3.8	0.22	0.35
three big, many small	0.05	100	51.7	3.8	0.25	0.40
gradient	0.05	100	27.5	1.2	0.20	0.37
equal k, unequal s (k=16)	0.05	100	86.8	2.8	0.17	0.42
equal k, unequal s (k=8)	0.05	100	88.7	0.3	0.02	0.06
equal s, unequal k	0.05	100	79.3	3.0	0.25	0.54
two worlds	0.10	200	117.6	11.7	0.43	0.75
three big, many small	0.10	200	74.6	8.8	0.47	0.81
gradient	0.10	200	34.3	5.0	0.53	0.75
equal k, unequal s (k=8)	0.10	200	153.1	2.5	0.16	0.42
equal k, unequal s (k=16)	0.10	200	154.6	9.6	0.50	0.95
equal s, unequal k	0.10	200	124.0	12.2	0.68	0.98

S3. FURTHER PROOFS AND DETAILS OF THE THEORY

This section completes Appendix A of the main text.

A. The two ends of the hierarchy

With the counts $c_{\mathcal{C}}(i,j)$ and pair fractions $\pi_{\mathcal{C}}(c)$ of the Omega index (Sec. VI D):

Lemma S1. *If $c_{\mathcal{C}}(i,j) = c$ for every pair and $c_{\mathcal{G}}$ is not identically c, then $\Omega = 0$.*

Proposition S1 (Endpoints)**.** *Let the ground truth give some pair count 0 and some pair count ≥ 1. (i) If G is connected and $|V| \geq 3$, then $\mathcal{C}_0 = \{V\}$ and $\Omega(0) = 0$. (ii) The communities at $t = 1$ are the classes of nodes with identical closed neighbourhoods having at least three members; if there are none, $\mathcal{C}_1 = \varnothing$ and $\Omega(1) = 0$. Consequently, when both hold, whenever $\Omega(t) > 0$ for some t the maximum $\Omega^\star = \max_t \Omega(t)$ is attained at an interior threshold $t^\star \in (0,1)$.*

The proposition does not assert that $\Omega^\star > 0$ or that $\Omega(t)$ has a single maximum; both are empirical. On noise-free ring lattices (ii) holds when $s \geq 2k+2$, and (i) when the merges connect all rings, which random merges usually do.

Proof of Lemma S1. $\pi_{\mathcal{C}}$ is concentrated on c, so $\Omega_e = \pi_{\mathcal{G}}(c) < 1$; a pair agrees exactly when $c_{\mathcal{G}}(i,j) = c$, so $\Omega_u = \pi_{\mathcal{G}}(c)$ as well. □

Proof of Proposition S1. (i) At $t = 0$ every wedge is admissible. Two edges sharing an endpoint form a wedge, and the wedges built on a common edge share its endpoints, so a walk in the line graph of G is a walk in L_0. The line graph of a connected graph is connected, and every node of a connected graph with $|V| \geq 3$ lies on a wedge. The single community V gives $c_{\mathcal{C}} \equiv 1$, and Lemma S1 applies. (ii) $J(u,v) = 1$ iff $N[u] = N[v]$, an equivalence relation whose classes are cliques; a triple is admissible at $t = 1$ iff it lies within one class, and triples within one class are connected in L_1. Without classes of size at least three, $c_{\mathcal{C}} \equiv 0$ and Lemma S1 applies. For the final claim: the cover changes only at the finitely many distinct triple scores, so $\Omega(t)$ takes finitely many values on $[0,1]$ and its maximum $\Omega^\star$ is attained; under the stated hypotheses $\Omega(0) = \Omega(1) = 0$ by (i) and (ii), so since $\Omega^\star \geq \Omega(t) > 0$ every maximiser lies in $(0,1)$. □

B. Further ring-lattice windows

Smaller rings and multiple merges. For a single merge between two rings, exact computation for $k = 4$ to 12 shows that the window of Eq. (4) also holds for ring sizes down to $s = k+4$ for $k = 4$ and 6, and $s = k+2$ for $k = 8$ to 12; smaller rings widen it upward (for $k = 4$ and $s = 6$ or 7 it is $(1/9, 2/5]$), which covers the rings of Fig. 2. A node formed by merging one node from each of m rings (all other merges isolated) gives the window $(1/(2k+1), \min\{k/(mk+2), \max((k-2)/(k+4), (k-1)/(mk+3))\}]$, whose upper edge tends to $1/m$ as k grows; it is non-empty if and only if $m \leq 2k$. These statements and Eqs. (4) and (5) were checked in rational arithmetic at and on either side of every distinct commonality value (analysis/exact_windows.py).

Unequal degrees. For one isolated merge between rings of even degrees $k_a, k_b \geq 4$ and sizes at least $2k_a + 2$ and $2k_b + 2$, the same argument applies with the commonalities of Lemma 1 for unequal degrees: with x, z on the ring of degree k_a, $J_u(d) = (k_a + 1 - d)/(k_a + k_b + 1 + d)$ and $J(z, y') = 1/(k_a + k_b + 1)$. The cross pair has $J = 1/(k_a + k_b + 1)$, which gives the lower edge of Eq. (5). On the ring of degree x, with y the degree of the other ring, the upper-edge argument runs with $J_u(1) = x/(x+y+2)$, $J_u(2) = (x-1)/(x+y+3)$ and $J(3) = (x-2)/(x+4)$, which gives $U(x,y)$; both rings must be recovered, which gives the minimum over $U(k_a, k_b)$ and $U(k_b, k_a)$.

Obstruction for unequal degrees. A shared node between a dense and a sparse community is held more weakly by the sparse one, so the window of such a merge narrows and moves down, with upper edge $k_b/(k_a + k_b + 2)$ for $k_a \gg k_b$. It cannot overlap the window of merges between two sparse rings, whose lower edge is $1/(2k_b + 1)$, once $k_a \geq 2k_b^2 - 2$: beyond that degree contrast no single threshold recovers a family that also contains merges between two sparse rings, even without noise.

C. Fusion of rings through shared pairs

Fusion of rings. Connectivity through shared pairs is what lets a community grow through its own open wedges, and Proposition 3 is its counterpart. On ring lattices, three communities that pairwise share single nodes are joined

when two of the three shared-node pairs are adjacent and the third lies within distance k; for $k = 8$ and pairwise adjacent shared nodes two apart, the rings are recovered only for $4/13 < t \leq 4/9$, and for $k = 4$ they are never recovered. Two rings joined only through two shared merged nodes, with no other merge within ring distance k of them, can be fused above the lower edge $1/(2k + 1)$ only when those nodes lie within ring distance k of each other on both rings, or within $k/2$ on one ring and within $3k/2$ on the other; for $k = 8$, shared nodes 2 apart on one ring and 11 apart on the other fuse the rings for $t \leq 3/23$. Theorem 2's isolation hypothesis excludes both cases. Whether fusion then occurs depends on the threshold, since the merge lowers the commonality of the shared pair: two rings with $s = 50$, $k = 8$ that share two nodes four positions apart are fused at small t but recovered separately at $t = 0.4$, where the shared pair, with $J = 1/3$, is not admitted.

D. Cost

Every commonality needed is available from the wedges themselves: for a pair $\{a, b\}$, $|N[a] \cap N[b]|$ is the number of common neighbours of a and b, that is of centred wedges a–c–b (a triangle contributes once to each of its three pairs), plus 2 if $a \sim b$, so one pass over the centred wedges, with pairs indexed by hashing, counts all of them. Our implementation finds components on the incidence graph between admitted triples and their pairs over a grid of thresholds. Alternatively, as in sequential clique percolation [8], a single union–find sweep over the triples in decreasing order of score builds the merge tree of the triple components at all thresholds in $O(W \log W)$, from which the cover at any one threshold is read in $O(W)$ more. For heavy-tailed degree distributions the hubs dominate through W, as for every triangle-based method.

Degree-ratio pruning. In the pruning of Sec. V C, every wedge through an edge with $\rho < t_{\min}$ is discarded. Common neighbours reached through discarded edges must still be counted, for example by intersecting the adjacency lists of the kept pairs; counting only the kept wedges would underestimate J. This is the size filter of exact set-similarity joins [9, 10], applied to closed neighbourhoods of sizes $d_u + 1$ and $d_v + 1$. The pruning applies to every rule that tests the edges of a wedge, including our three ablations, but not to link communities, which score a wedge by its non-adjacent pair alone, whose commonality is not bounded by the degrees of the wedge's edges, so every wedge through a hub has to be enumerated. Our implementation counts the common neighbours of the kept pairs from the full adjacency matrix, and pruned and unpruned runs give identical covers on its test graphs, which include a hub joined to many leaves, and on the full Amazon and DBLP graphs at all 30 thresholds $t \geq 0.02$ of our grid.

S4. COMMONALITY IN EMPIRICAL NETWORKS

Commonality in empirical networks: details

Unlabelled nodes. Edges touching unlabelled DBLP authors join low-degree nodes in small, tightly knit groups – the median larger endpoint degree of the pooled "different" class is 8, against 22 for same-venue edges – and have high commonality (median $J = 0.40$); pooled into the "different" class they push its median above that of same-venue pairs, and the AUC for separating the classes falls to 0.41 [Fig. S1(f)]. Even after unlabelled edges are set apart, "outer" means no recorded shared group, not proven separation: on the same DBLP co-authorship network with venue groups, and on an Amazon co-purchasing network with product-category groups, standard methods recover the recorded groups poorly [11]. With unlabelled edges set apart, the inner and outer medians of closed-neighbourhood Jaccard are 0.29 and 0.15 in the karate club, 0.25 and 0.17 in DBLP, 0.27 and 0.20 in Amazon, and 0.032 and 0.022 in YouTube [Fig. S1(a–d)].

Degree strata. The stratum AUCs of Amazon and YouTube lie around their pooled 0.60 and 0.61 [Fig. S1(g,h)]. Their weaker signal is consistent with what their ground truth is: Amazon's product categories are nested, with 95% of labelled products in several categories, and YouTube's groups are user-declared interests rather than structural units [12].

The family of functions. With $I = |N[u] \cap N[v]|$ and $U = |N[u] \cup N[v]|$, the eight functions of Fig. S2 are the raw overlap I; Jaccard I/U [13]; the generalisations I^2/U and $I/U^{1/2}$ of the form I^a/U^b (the first ranks pairs as the degree-normalised score $I^2/(\langle d \rangle U)$ of Ref. [14], and the two rank pairs identically, since $I/U^{1/2} = (I^2/U)^{1/2}$, so they count as one choice for any threshold-based use); the cosine $I/\sqrt{|N[u]||N[v]|}$ [15] used by SCAN [16]; the overlap normalised by the smaller and by the larger neighbourhood, $I/\min$ and $I/\max$ (closed-neighbourhood versions of the hub-promoted and hub-depressed indices [17, 18], the first derived from the topological overlap of Ravasz *et al.* [19]); and, as a structurally different score, resource allocation [17] $\sum_{w \in N(u) \cap N(v)} 1/d_w$, which weights shared neighbours by their rarity. The karate club is too small to stratify; there every function separates the classes in aggregate, with AUC 0.73–0.89. That the overlap-based scores perform alike within strata is expected, since the strata are narrow degree bins. Commonality is therefore best understood as a family whose appropriate member depends on the network.

Pairs and triples. We sample wedges whose three nodes are labelled and weight closed triangles by 1/3, so that the population is unique triples rather than centred wedges. The pair rule tests two pairs for an open wedge and three for a triangle; Fig. S3 compares the two rules over the full range of thresholds. At recall 0.2, 91% of pair-admitted triples are pure against 92% of triple-admitted ones in Amazon (base rate 86%), and 10% against 13% in YouTube (base rate 10%). Closed triangles alone are 99% pure in DBLP, reflecting cliques of co-authors, but cover only a quarter of the pure triples; open wedges supply the rest, which is the trade-off the triple rule manages.

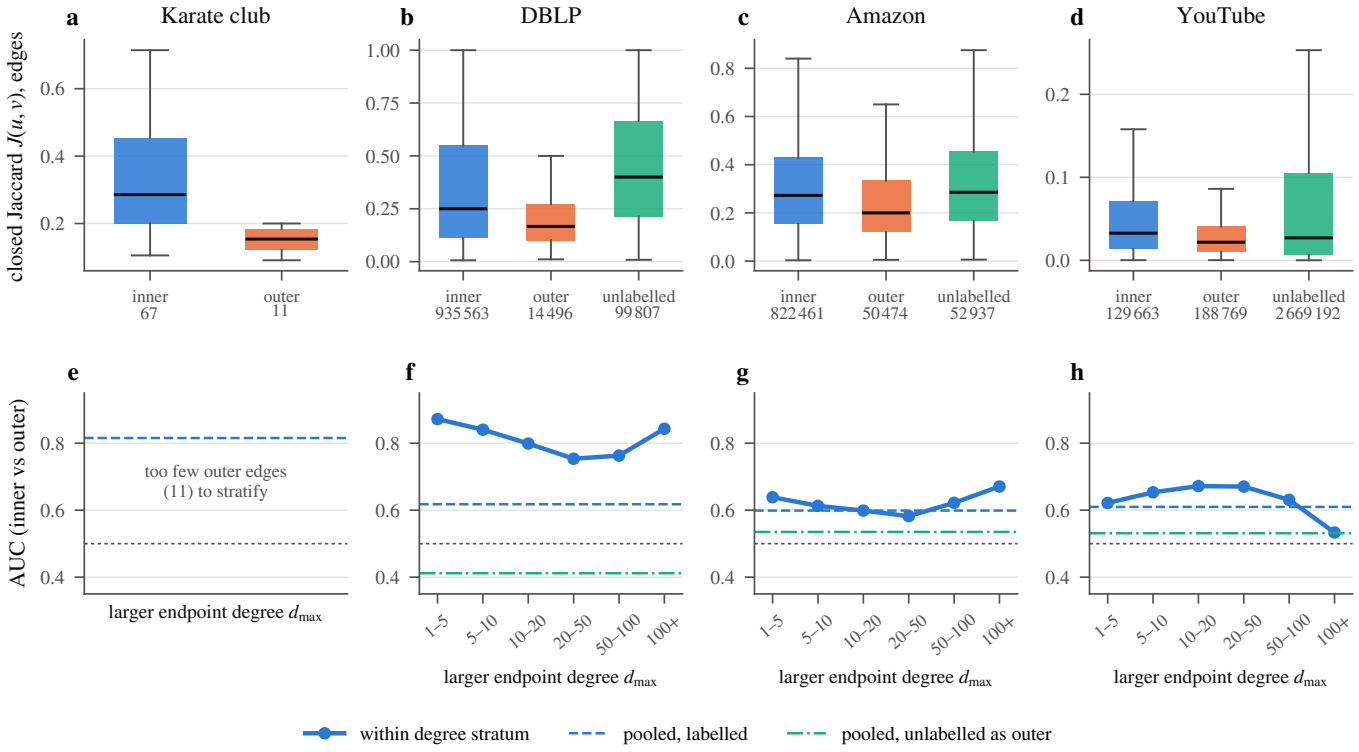


FIG. S1. Commonality of edges in four networks with recorded memberships. (a–d) Closed-neighbourhood Jaccard of edges whose endpoints share a recorded group (inner), are both labelled but share none (outer), or include an unlabelled node; boxes span the quartiles, whiskers the 5th–95th percentiles, numbers give edge counts. (e–h) AUC for separating inner from outer edges within strata of the larger endpoint degree (points), pooled over all labelled edges (dashed), and with unlabelled edges counted as outer (dash-dotted), as in the earlier analysis of Ref. [14].

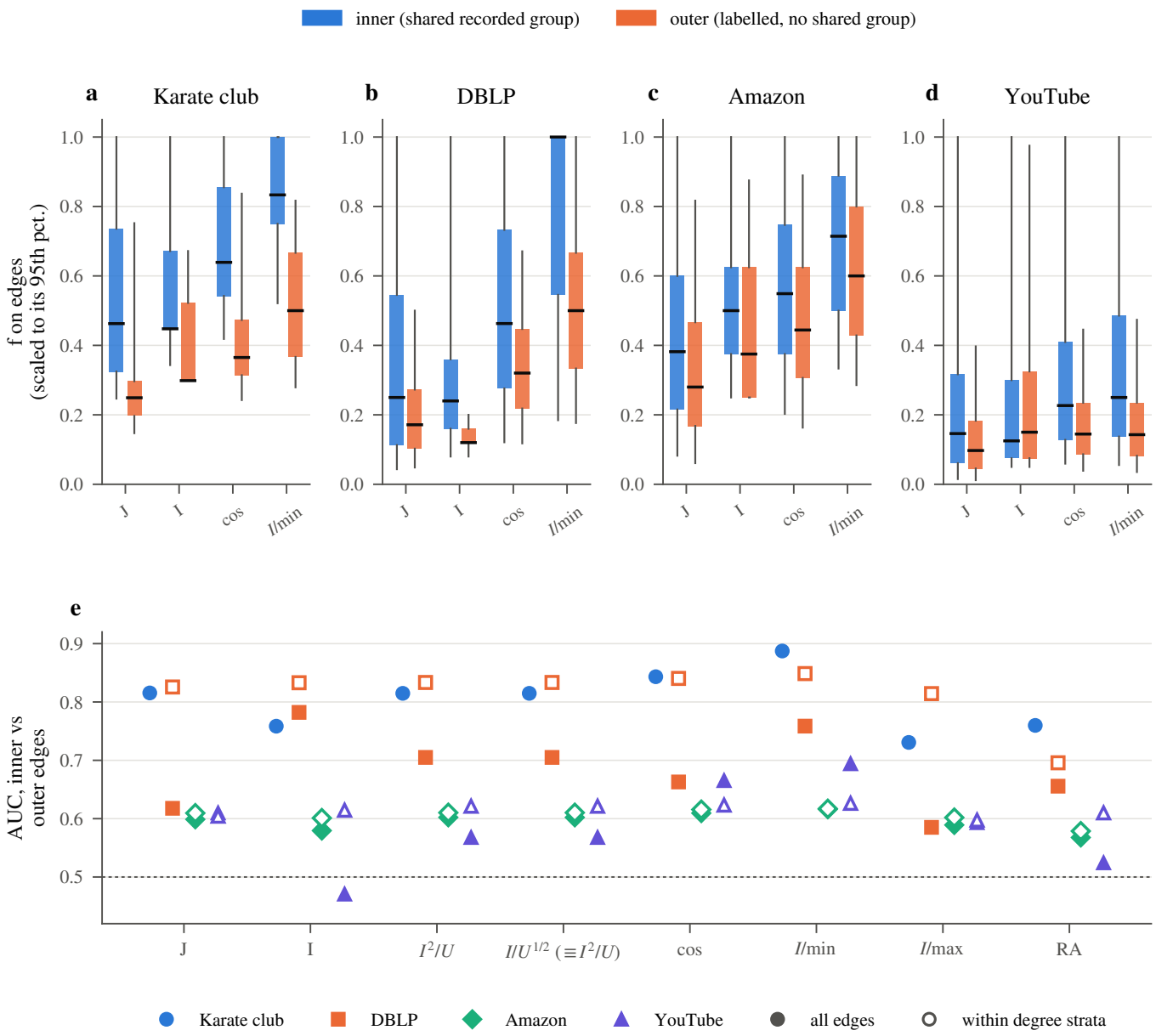


FIG. S2. Commonality as a family of functions. (a–d) Inner (shared recorded group) and outer (labelled, no shared group) edges under four functions, each scaled to its own 95th percentile. (e) AUC for separating inner from outer edges with eight scores, over all labelled edges (filled) and within strata of both endpoint degrees (open); $I/U^{1/2}$ ranks pairs as I^2/U does. The karate club is too small to stratify.

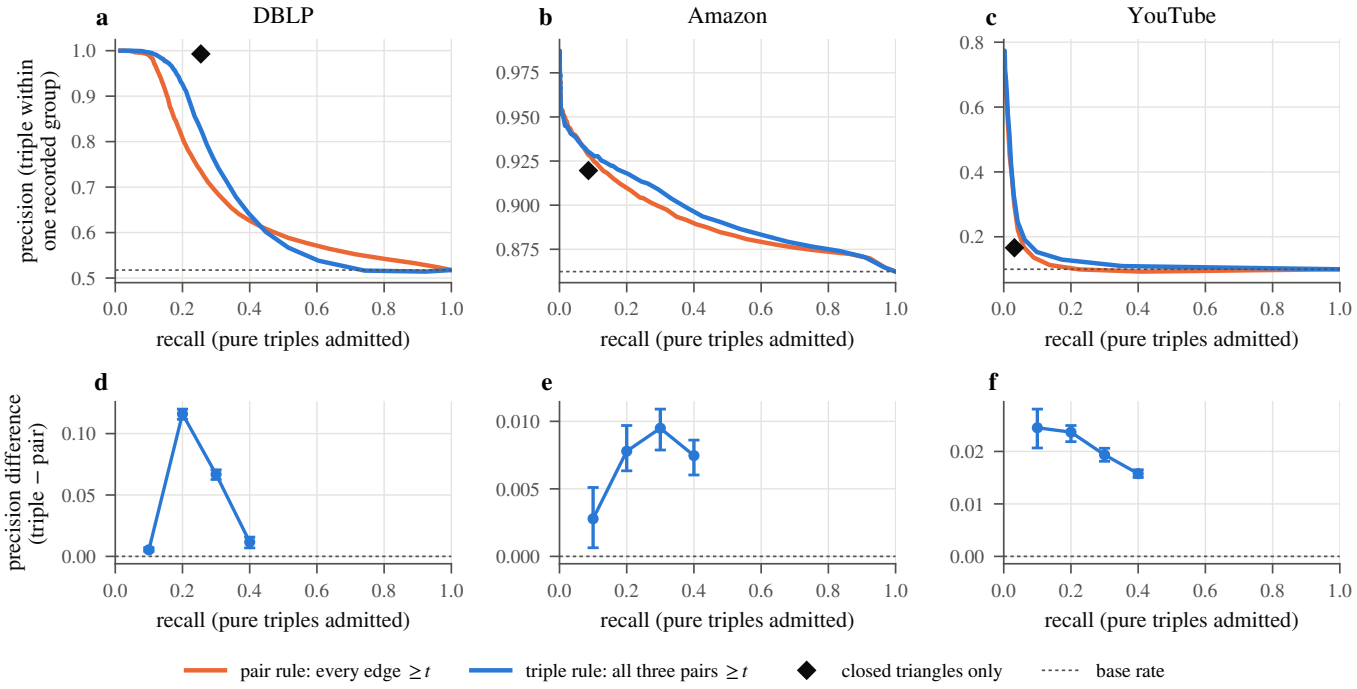


FIG. S3. Pair rule versus triple rule on unique labelled triples (wedges sampled uniformly, triangles weighted 1/3). Precision is the fraction of admitted triples contained in one recorded group, recall the fraction of such triples admitted, as the threshold varies. Diamond: closed triangles only, without a threshold. Bottom row: difference in precision between the triple and the pair rule at equal recall, with 95% intervals from resampling the triples; the y-axes of the top row differ between panels.

TABLE S6. Accuracy on the benchmarks as a user would run the methods, every parameter chosen without labels by one rule: on each test graph, the parameter whose cover has the largest overlapping modularity, Eq. (7), which for a partition is the Newman–Girvan modularity (label-free protocol). Ω, mean over the noise range ($p \in [0, 0.8]$ for rings, $\mu \in [0.1, 0.6]$ for LFR), ten test graphs per noise level. HC: HyperCommon; e-c.: its triples joined only through shared edges; tri.: triangles only (J-thresholded 3-clique percolation [8, 20]); 2-test: third pair not tested; Link: link communities with the same similarity; CPM: clique percolation; Leiden: Leiden with the Reichardt–Bornholdt modularity; L-CPM: Leiden with the Constant Potts Model; Info.: Infomap; Walk.: Walktrap, walks of length 4, dendrogram cut at maximum modularity; LPA: label propagation; SBM: nested degree-corrected stochastic block model, each level of its hierarchy a parameter value; OSLOM: order statistics local optimisation, significance threshold t as its parameter; SCAN: SCAN with cosine similarity; SCAN-J: SCAN with Jaccard similarity; α: exponent of the preferential (degree-biased) noise. Parameter-free methods (Walktrap, LPA) are as under every protocol. Superscripts on HC: the paired 95% bootstrap interval of its difference from the best other method (ablations excluded) lies above zero (+), contains zero (=) or lies below zero (−). Bold: best in row, within 0.005. 95% bootstrap half-widths of these means over the ten test seeds have median 0.007 and maximum 0.050.

		HC and ablations				relatives		partitions						overlapping methods					
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	OSLOM	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	**0.61**$^{=}$	**0.61**	0.51	0.51	0.60	0.51	**0.61**	**0.61**	0.48	0.39	0.14	0.38	0.36	0.55	0.53	0.52	0.50	0.53
	0.05	**0.55**$^{+}$	0.54	0.49	0.29	0.53	0.49	0.44	0.43	0.32	0.24	0.12	0.27	0.33	0.45	0.44	0.39	0.26	0.30
	0.10	**0.50**$^{+}$	**0.50**	0.46	0.22	0.45	0.44	0.32	0.31	0.25	0.18	0.11	0.21	0.30	0.37	0.40	0.40	0.21	0.21
uniform, k=16	0.00	0.78^{-}	0.78	0.75	0.69	0.76	0.67	**0.84**	0.84	0.47	0.65	0.14	0.50	0.34	0.72	0.62	0.67	0.65	0.59
	0.05	0.72^{-}	0.71	0.69	0.39	0.54	0.56	**0.75**	0.75	0.35	0.47	0.11	0.38	0.32	0.64	0.50	0.63	0.33	0.35
	0.10	**0.62**$^{=}$	**0.62**	0.60	0.29	0.39	0.54	**0.62**	0.61	0.29	0.33	0.09	0.28	0.29	0.54	0.45	0.57	0.24	0.25
equal k, unequal s, k=8	0.00	**0.57**$^{+}$	0.57	0.46	0.49	0.56	0.46	0.56	0.56	0.36	0.33	0.10	0.33	0.25	0.47	0.48	0.49	0.47	0.49
	0.05	**0.52**$^{+}$	0.51	0.45	0.25	0.49	0.43	0.35	0.34	0.24	0.19	0.09	0.23	0.23	0.34	0.38	0.35	0.22	0.23
	0.10	**0.45**$^{+}$	**0.45**	0.40	0.17	0.41	0.37	0.24	0.24	0.19	0.15	0.08	0.16	0.21	0.27	0.33	0.33	0.15	0.16
equal k, unequal s, k=16	0.00	0.80^{-}	0.79	0.77	0.70	0.77	0.71	**0.84**	**0.84**	0.60	0.69	0.17	0.49	0.43	0.76	0.64	0.69	0.66	0.60
	0.05	**0.73**$^{=}$	**0.73**	0.71	0.42	0.56	0.59	**0.73**	0.72	0.51	0.56	0.14	0.40	0.41	0.70	0.51	0.66	0.37	0.37
	0.10	**0.65**$^{=}$	**0.65**	0.62	0.31	0.42	0.57	0.60	0.59	0.44	0.44	0.12	0.34	0.37	0.64	0.45	0.61	0.27	0.29
equal s, unequal k	0.00	0.78$^{=}$	**0.78**	0.76	0.71	0.74	0.60	**0.79**	**0.79**	0.55	0.60	0.16	0.51	0.35	0.56	0.53	0.64	0.66	0.61
	0.05	0.70$^{=}$	**0.70**	0.69	0.47	0.57	0.59	0.67	**0.70**	0.44	0.52	0.13	0.40	0.31	0.50	0.38	0.57	0.40	0.41
	0.10	0.61^{-}	0.61	0.59	0.39	0.45	0.56	0.58	**0.63**	0.39	0.46	0.11	0.34	0.27	0.46	0.29	0.46	0.32	0.34
two worlds	0.00	**0.83**$^{+}$	**0.82**	0.82	0.71	0.78	0.70	0.58	0.79	0.40	0.52	0.13	0.27	0.28	0.31	0.48	0.65	0.72	0.64
	0.05	**0.76**$^{+}$	**0.76**	**0.76**	0.58	0.66	0.66	0.44	0.64	0.29	0.41	0.10	0.22	0.25	0.27	0.35	0.55	0.51	0.49
	0.10	**0.71**$^{+}$	**0.71**	0.70	0.49	0.53	0.65	0.38	0.50	0.26	0.32	0.08	0.18	0.22	0.23	0.28	0.43	0.42	0.43
three big, many small	0.00	**0.83**$^{+}$	**0.83**	0.82	0.76	0.81	0.72	0.33	0.40	0.21	0.27	0.07	0.18	0.15	0.18	0.47	0.64	0.74	0.66
	0.05	**0.78**$^{+}$	**0.78**	0.74	0.48	0.67	0.68	0.28	0.34	0.16	0.22	0.06	0.13	0.14	0.16	0.35	0.51	0.42	0.44
	0.10	**0.67**$^{+}$	0.66	0.62	0.36	0.55	0.61	0.23	0.29	0.14	0.17	0.05	0.10	0.12	0.14	0.29	0.43	0.33	0.40
gradient	0.00	0.42^{-}	0.40	0.39	0.36	**0.61**	0.56	0.21	0.21	0.11	0.14	0.03	0.14	0.07	0.16	0.50	0.48	0.41	0.38
	0.05	0.32^{-}	0.31	0.27	0.20	**0.48**	0.40	0.14	0.15	0.07	0.10	0.03	0.07	0.07	0.11	0.34	0.42	0.18	0.16
	0.10	0.23^{-}	0.24	0.22	0.13	**0.39**	0.35	0.09	0.11	0.06	0.07	0.02	0.05	0.06	0.09	0.27	0.33	0.12	0.13
uniform k=8, α=1	0.05	**0.56**$^{+}$	0.55	0.49	0.31	0.53	0.48	0.45	0.44	0.32	0.24	0.12	0.28	0.33	0.46	0.43	0.37	0.28	0.32
uniform k=8, α=2	0.05	**0.56**$^{+}$	**0.56**	0.49	0.33	0.54	0.48	0.45	0.45	0.32	0.22	0.12	0.28	0.33	0.46	0.43	0.39	0.30	0.32
two worlds, α=1	0.05	**0.77**$^{+}$	**0.77**	0.75	0.54	0.66	0.65	0.47	0.66	0.31	0.62	0.11	0.26	0.26	0.28	0.26	0.44	0.46	0.47
two worlds, α=2	0.05	**0.79**$^{+}$	**0.79**	0.75	0.51	0.66	0.64	0.49	0.66	0.32	0.62	0.10	0.29	0.26	0.30	0.23	0.41	0.50	0.50
LFR, o_n=0		0.84^{-}	0.86	0.84	0.56	0.85	0.66	0.98	0.98	**1.00**	0.98	0.61	**0.99**	0.91	**1.00**	0.46	0.77	0.43	0.39
LFR, o_n=0.1		0.69^{-}	0.72	0.71	0.27	0.53	0.60	0.78	0.78	0.88	0.81	0.49	0.87	0.79	**0.94**	0.41	0.72	0.29	0.23
LFR, o_n=0.3		0.45^{-}	0.50	0.46	0.11	0.31	0.48	0.54	0.53	0.65	0.49	0.32	0.46	0.60	**0.85**	0.40	0.64	0.13	0.10
LFR, $\langle k \rangle$=10		0.43^{-}	0.45	0.35	0.09	0.40	0.36	0.69	0.70	0.70	0.61	0.36	0.63	0.48	**0.77**	0.26	0.25	0.12	0.12
LFR, o_m=4		0.61^{-}	0.63	0.62	0.40	0.46	0.52	0.62	0.62	0.70	0.62	0.40	0.67	0.68	**0.75**	0.43	0.66	0.36	0.28
LFR, sizes 50–250		0.42^{-}	0.41	0.42	0.14	0.40	0.16	0.87	0.85	0.79	0.75	0.38	0.79	0.59	**0.89**	0.18	0.55	0.13	0.07
LFR, C_{LFR}=0.3		0.33^{-}	0.33	0.32	0.19	0.17	0.39	0.50	0.50	0.54	0.52	0.48	0.38	**0.56**	0.54	0.18	0.30	0.20	0.20
LFR, C_{LFR}=0.5		0.24^{-}	0.24	0.24	0.19	0.10	0.24	0.29	0.29	0.30	0.26	**0.35**	0.24	**0.35**	0.33	0.17	0.21	0.18	0.20

S5. BENCHMARKS

This section gives the full benchmark tables behind Sec. VII and the details the main text summarises. Numbers quoted for a ring family without its overlap level are at $o_n = 0.05$.

Recovery window and fusions. At $o_n = 0.05$, for $k = 16$ the isolation hypothesis of Theorem 2 cannot hold, and for $k = 8$ it would need exactly five merge positions more than k apart on every ring. Rings are then fused inside the window in three ways: two rings share two merged nodes within ring distance k on both rings (Proposition 3); they share two merged nodes within $k/2$ on one ring and within $3k/2$ on the other; or the merged nodes of three pairwise-merged rings form an admissible wedge. Without noise, between 6% and 61% of the rings of a family are fused at $o_n = 0.05$ and between 38% and all of them at $o_n = 0.10$ (Table S5). For $k = 16$ the noise-free curve of

TABLE S7. Robustness to noise on the ring families: mean Ω over the high-noise half ($p \in [0.4, 0.8]$) and $p_{1/2}$, the rewired fraction at which the mean Ω over the ten test graphs first falls below 0.5 (linear interpolation; > 0.8: never), for HyperCommon (HC) and the best of the other methods by each measure, under the oracle and label-free protocols. Bold: HC ahead or within 0.005 (0.01 for $p_{1/2}$).

	oracle				label-free			
Family	HC $\bar{\Omega}_{\text{hi}}$	best other	HC $p_{1/2}$	best other	HC $\bar{\Omega}_{\text{hi}}$	best other	HC $p_{1/2}$	best other
uniform, k=8, o_n=0	0.29	0.39 (SLPA)	0.48	0.52 (Leiden)	0.28	0.30 (SLPA)	**0.48**	0.48 (L-CPM)
uniform, k=8, o_n=0.05	0.25	0.35 (SLPA)	**0.46**	0.45 (Link)	0.24	0.27 (SLPA)	**0.45**	0.45 (Link)
uniform, k=8, o_n=0.10	0.22	0.31 (SLPA)	**0.42**	0.42 (Link)	0.21	0.24 (SLPA)	**0.41**	0.38 (Link)
uniform, k=16, o_n=0	0.56	0.72 (Leiden)	0.61	0.68 (Leiden)	0.56	0.69 (Leiden)	0.61	0.67 (Leiden)
uniform, k=16, o_n=0.05	0.47	0.59 (Leiden)	0.58	0.64 (Leiden)	0.46	0.56 (Leiden)	0.58	0.63 (Leiden)
uniform, k=16, o_n=0.10	0.37	0.44 (Leiden)	0.54	0.58 (Leiden)	0.36	0.40 (Leiden)	0.53	0.57 (Leiden)
equal k, unequal s (k=8)	0.19	0.26 (SLPA)	**0.41**	0.40 (Link)	**0.19**	0.19 (SLPA)	**0.41**	0.40 (Link)
equal k, unequal s (k=16)	0.51	0.63 (Leiden)	0.60	0.68 (Leiden)	0.49	0.54 (Leiden)	0.59	0.64 (Leiden)
equal s, unequal k	0.53	0.60 (L-CPM)	0.66	0.69 (L-CPM)	0.52	0.55 (L-CPM)	**0.66**	0.66 (L-CPM)
two worlds	0.61	0.62 (L-CPM)	**0.68**	0.66 (Link)	**0.59**	0.52 (Link)	**0.68**	0.65 (Link)
three big, many small	**0.63**	0.54 (Link)	**0.67**	0.65 (Link)	**0.60**	0.49 (Link)	**0.66**	0.62 (Link)
gradient of ten rings	0.23	0.26 (Link)	0.47	0.48 (Link)	0.14	0.18 (Link)	0.17	0.41 (Link)
uniform k=8, pref. α=1	0.25	0.35 (SLPA)	**0.46**	0.46 (Leiden)	0.24	0.27 (SLPA)	**0.46**	0.45 (Link)
uniform k=8, pref. α=2	0.25	0.34 (SLPA)	**0.46**	0.46 (Leiden)	0.24	0.27 (SLPA)	**0.46**	0.45 (Link)
two worlds, pref. α=1	0.60	0.70 (L-CPM)	0.67	0.70 (L-CPM)	**0.59**	0.50 (L-CPM)	0.67	0.69 (L-CPM)
two worlds, pref. α=2	0.61	0.64 (L-CPM)	**0.67**	0.66 (L-CPM)	**0.60**	0.48 (L-CPM)	**0.67**	0.62 (L-CPM)

Fig. 3(b) reaches 0.95 only for $t \geq 0.30$.

Label-free and oracle protocols. Table S6 gives the label-free Ω of every method and Table S8 the oracle one. Under superlinear preferential rewiring HyperCommon reaches 0.79 on two worlds against 0.66 for the Constant Potts Model and link communities, and on uniform rings with $k = 8$ it leads with overlap (0.50 against 0.45 for link communities at $o_n = 0.10$). With labels, Leiden with one objective or the other is ahead on most rings without overlap, on the dense rings at $o_n = 0.05$ and on rings of equal size and unequal degree; the Constant Potts Model is ahead on two worlds at $o_n \leq 0.05$ by at most 0.05. On the gradient at $p = 0$, the resolution at which Leiden returns the right number of communities gives $\Omega = 0.26$, and HyperCommon attains 0.97 at the threshold at which it returns ten. Under the native protocol HyperCommon is behind on the dense rings with little overlap, where Leiden at $\gamma = 1$ leads, and on the gradient; link communities lose most from their own rule, the cut of maximum partition density returning 640–1320 communities. The frozen and native tables are in the repository.

Per-community recovery and ONMI. On the two families with a few large dense rings and many small sparse ones, at the best threshold for the large rings part of each small sparse ring has commonality below it, and HyperCommon splits such rings and leaves part of their nodes unassigned (coverage 87%–88%). By ONMI the Constant Potts Model is then ahead on two worlds (0.57 against 0.42) and on three big, many small (0.52 against 0.40), where HyperCommon leads by Ω. On the label-free covers, with its resolution chosen by modularity, the Constant Potts Model recovers the small rings less well and its ONMI lead on these two families is gone at $o_n = 0.05$.

Robustness to noise

Table S7 reports, for every ring family, the rewired fraction $p_{1/2}$ at which the mean Ω first falls below one half and the mean Ω over the high-noise half, $p \in [0.4, 0.8]$. Without labels, on the four families whose large rings have degree 32 and small rings degree 8, HyperCommon has the highest high-noise mean (0.60 against 0.49 for link communities on three big, many small); on the gradient it is the least robust of the leading methods. Only DEMON and ANGEL degrade under degree-biased noise; for HyperCommon the bound of Jaccard by the degree ratio offers a mechanism: a hub grown by preferential rewiring has low commonality with the lattice neighbours it attaches to, so the edges it gains rarely pass.

Ablations and the commonality function

Without the third-pair test, a wedge whose two edges pass reaches into another community through a shared node or through an edge between communities, which is how the two-test variant loses 0.21 even on LFR graphs without overlap. Triangles alone are better than HyperCommon in two settings under the oracle protocol and in none under the label-free one; with one parameter frozen per family they have the higher mean Ω in 15 of the 36 settings. Joining triples only through shared edges is slightly better on the LFR graphs, by up to 0.03 with labels and 0.05 without.

Tables S17 and S18 run HyperCommon with each commonality function on eighteen settings (six uniform-ring, six unequal-ring at $o_n = 0.05$ and six LFR), the threshold taken as a quantile of each function's own triple scores. With the best threshold Jaccard is the best, or within 0.01, on seventeen and never more than 0.02 behind. The two unnormalised scores, the raw overlap and resource allocation, are less accurate than Jaccard on all eighteen, by up to 0.21 and 0.20: where communities differ in degree, a threshold on an unnormalised score is a threshold on degree as much as on commonality. Choosing the function together with the threshold raises Ω by 0.01–0.06. Without labels, on the gradient the selected cover of I^2/U reaches $\Omega = 0.50$ against 0.36 for Jaccard, but the selector, asked to choose the function as well, does not find it (0.31).

LFR graphs, geometry, controls and null graphs

With its best threshold HyperCommon reaches $\Omega = 0.885$ at $\mu = 0.1$ and $o_n = 0.1$, against 0.884 for Leiden with modularity and for Infomap and 0.886 for Leiden with the Constant Potts Model, while SLPA reaches 0.99 [Fig. S4(f)]. At $\mu = 0.3$ only 5% of the wedges inside communities pass at $t = 0.1$ in the slice with communities of 50 to 250 nodes. The generator's own rewiring, used to raise the clustering, adds triangles across communities as well: the share of triangles spanning two communities grows from 2% to 19% and 32%.

In the geometry families the mean clustering of the noiseless graphs falls from 0.62–0.68 at $q = 0$ to 0.48–0.51 at $q = 0.1$ and 0.04–0.16 at $q = 1$ (Table II). On lattice communities HyperCommon leads with labels by 0.94 against 0.87 for clique percolation on three big, many small and is level with the Constant Potts Model on two worlds (0.91); with labels at $q = 1$ on the uniform rings it reaches 0.60 against 0.85. Open wedges slow the decline: there triangles alone fall from 0.65 to 0.29 as q goes from 0 to 1, and HyperCommon from 0.73 to 0.60.

Among the controls (Table S15), with hubs inside the communities the Leiden objectives gain and lead, Leiden on the dense hub rings (0.83 against 0.69 without labels) and the Constant Potts Model on two worlds with hubs (0.83 against 0.75). With split overlap HyperCommon is level with the best other method or ahead on two worlds and three big, many small under both protocols. With random geometric communities it is behind Leiden with labels (0.50 against 0.57) and only slightly ahead of it without labels on the uniform rings (0.48 against 0.46), and behind the Constant Potts Model on two worlds under both protocols (0.73 against 0.84 and 0.79). On the null graphs (Table S16) label-free HyperCommon covers 98% and 96% of the nodes with its 101 and 175 communities, against 32–36 and 12–28 for Leiden, Infomap, OSLOM and the stochastic block model.

Real networks

On the SNAP graphs, Leiden's label-free F1 equals its F1 at $\gamma = 1$ (Table III). With labels Infomap leads by F1 on Amazon (0.935), OSLOM on DBLP (0.549, against 0.545 for Infomap) and Leiden on YouTube (0.363); Infomap at its default Markov time is about 0.06 ahead of label-free HyperCommon on Amazon and DBLP. On Amazon, whose communities are sparse in triangles, the two-test variant reaches F1 0.887 but a lower Ω (0.424 against 0.459). Detecting on the subgraphs induced by the labelled nodes, as is common, makes F1 uninformative on Amazon and YouTube: there the connected components score within 0.005 of the best method.

On the fourteen smaller networks (Tables S20, S21, S22 and S23), HC_f chooses, with labels, among 700 (function, threshold) pairs per network, an upper bound rather than one parameter grid; it is the most accurate method on four networks by F1 and four by Ω. OSLOM has the best mean rank without labels (6.2) and Leiden with labels (6.0); OSLOM, Leiden, SLPA, label propagation, SCAN and Infomap each lead on some networks. The label-free threshold retains 95% of the best F1 or more on seven of the fourteen; on political blogs, a two-group network in which HyperCommon's best cover is the giant component, it selects fifteen small communities instead.

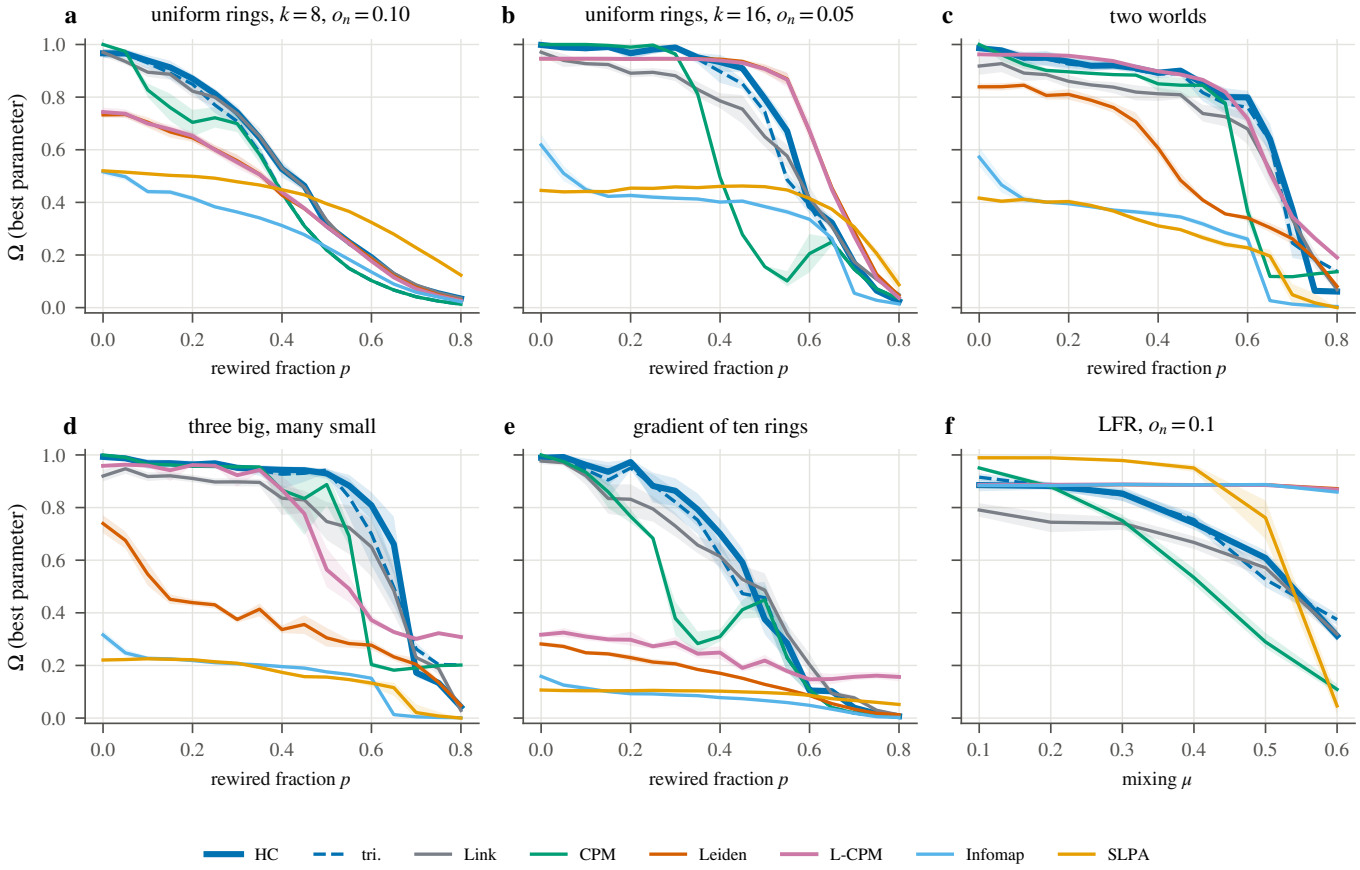


FIG. S4. As Fig. 4, with every method at its best parameter on each graph (oracle protocol).

TABLE S8. As Table S6, with every method at its best parameter on each test graph (oracle protocol). 95% bootstrap half-widths of these means over the ten test seeds have median 0.005 and maximum 0.049. Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods					
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	OSLOM	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.62$^-$	0.62	0.51	0.55	0.62	0.51	**0.68**	**0.67**	0.49	0.39	0.14	0.42	0.47	0.56	0.54	0.54	0.54	0.54
	0.05	**0.57**$^+$	**0.57**	0.50	0.34	**0.57**	0.49	0.55	0.55	0.35	0.24	0.12	0.32	0.44	0.46	0.44	0.43	0.30	0.31
	0.10	**0.52**$^+$	0.52	0.47	0.25	0.52	0.45	0.41	0.41	0.28	0.18	0.11	0.25	0.40	0.38	0.40	0.41	0.22	0.22
uniform, k=16	0.00	0.78$^-$	0.78	0.75	0.70	0.78	0.68	**0.86**	0.85	0.47	0.65	0.14	0.51	0.44	0.73	0.64	0.67	0.65	0.59
	0.05	0.73$^-$	0.72	0.70	0.42	0.66	0.56	**0.77**	**0.77**	0.35	0.47	0.11	0.38	0.41	0.66	0.51	0.63	0.34	0.36
	0.10	0.63$^=$	0.63	0.61	0.31	0.53	0.54	**0.64**	0.63	0.29	0.33	0.09	0.29	0.38	0.58	0.45	0.57	0.24	0.26
equal k, unequal s, k=8	0.00	0.58$^-$	0.57	0.46	0.51	0.57	0.46	**0.60**	**0.60**	0.36	0.33	0.10	0.33	0.35	0.48	0.49	0.49	0.49	0.49
	0.05	**0.52**$^+$	0.52	0.45	0.27	0.52	0.43	0.40	0.39	0.25	0.19	0.09	0.24	0.32	0.35	0.39	0.35	0.23	0.24
	0.10	**0.46**$^=$	0.45	0.41	0.19	**0.46**	0.37	0.29	0.29	0.20	0.15	0.08	0.18	0.29	0.28	0.35	0.34	0.16	0.17
equal k, unequal s, k=16	0.00	0.80$^-$	0.79	0.77	0.72	0.79	0.71	**0.87**	**0.87**	0.61	0.69	0.17	0.50	0.53	0.77	0.67	0.70	0.66	0.60
	0.05	0.75$^-$	0.74	0.72	0.47	0.69	0.59	**0.79**	0.78	0.51	0.56	0.14	0.42	0.50	0.71	0.55	0.66	0.39	0.39
	0.10	**0.67**$^+$	**0.67**	0.65	0.35	0.59	0.57	0.66	0.65	0.44	0.44	0.12	0.36	0.46	0.65	0.49	0.62	0.28	0.29
equal s, unequal k	0.00	0.79$^-$	0.79	0.77	0.73	0.78	0.61	**0.82**	**0.82**	0.55	0.60	0.16	0.52	0.40	0.58	0.54	0.64	0.68	0.61
	0.05	0.71$^-$	0.72	0.70	0.49	0.64	0.59	**0.72**	**0.72**	0.45	0.52	0.13	0.42	0.35	0.52	0.41	0.58	0.41	0.42
	0.10	0.62$^-$	0.62	0.60	0.41	0.53	0.56	0.63	**0.65**	0.39	0.46	0.11	0.35	0.31	0.47	0.34	0.49	0.32	0.35
two worlds	0.00	**0.84**$^-$	**0.83**	0.82	0.75	0.82	0.70	0.67	**0.84**	0.40	0.52	0.13	0.29	0.32	0.36	0.53	0.66	0.72	0.64
	0.05	0.78$^-$	0.78	0.77	0.59	0.71	0.68	0.56	**0.78**	0.29	0.41	0.10	0.24	0.28	0.32	0.40	0.58	0.51	0.49
	0.10	**0.72**$^=$	**0.72**	0.71	0.50	0.61	0.66	0.45	**0.72**	0.26	0.32	0.08	0.20	0.24	0.28	0.35	0.49	0.43	0.44
three big, many small	0.00	**0.84**$^+$	**0.84**	0.84	0.77	0.83	0.74	0.49	0.81	0.24	0.27	0.07	0.24	0.18	0.20	0.50	0.65	0.75	0.67
	0.05	**0.80**$^+$	**0.80**	0.79	0.57	0.72	0.71	0.37	0.71	0.16	0.22	0.06	0.16	0.16	0.18	0.38	0.53	0.49	0.50
	0.10	0.71$^+$	**0.72**	0.69	0.48	0.61	0.67	0.26	0.48	0.14	0.17	0.05	0.13	0.13	0.16	0.33	0.45	0.42	0.44
gradient	0.00	0.65$^-$	0.64	0.60	0.57	**0.66**	0.57	0.35	0.44	0.14	0.14	0.03	0.18	0.11	0.21	0.54	0.54	0.54	0.51
	0.05	**0.57**$^+$	**0.57**	0.54	0.29	0.54	0.43	0.16	0.23	0.07	0.10	0.03	0.08	0.09	0.15	0.43	0.47	0.22	0.23
	0.10	**0.48**$^+$	**0.48**	0.47	0.25	0.45	0.41	0.10	0.18	0.06	0.07	0.02	0.06	0.08	0.11	0.37	0.37	0.16	0.18
uniform k=8, α=1	0.05	**0.58**$^+$	0.57	0.50	0.35	0.57	0.49	0.56	0.55	0.35	0.24	0.12	0.32	0.44	0.46	0.44	0.43	0.31	0.33
uniform k=8, α=2	0.05	**0.58**$^+$	**0.58**	0.50	0.38	0.57	0.48	0.56	0.55	0.35	0.22	0.12	0.32	0.44	0.46	0.44	0.43	0.33	0.36
two worlds, α=1	0.05	0.78$^-$	0.78	0.76	0.56	0.70	0.67	0.61	**0.82**	0.31	0.62	0.11	0.27	0.30	0.33	0.35	0.47	0.47	0.47
two worlds, α=2	0.05	**0.79**$^=$	**0.79**	0.77	0.57	0.72	0.65	0.66	**0.79**	0.34	0.62	0.10	0.30	0.29	0.35	0.32	0.44	0.51	0.52
LFR, o_n=0		0.86$^-$	0.86	0.85	0.65	0.85	0.66	**1.00**	**1.00**	**1.00**	0.98	0.61	0.99	0.91	**1.00**	0.51	0.77	0.57	0.41
LFR, o_n=0.1		0.74$^-$	0.75	0.73	0.34	0.66	0.60	0.88	0.88	0.88	0.81	0.49	0.87	0.84	**0.97**	0.46	0.73	0.33	0.25
LFR, o_n=0.3		0.50$^-$	0.53	0.51	0.16	0.42	0.48	0.69	0.69	0.66	0.49	0.32	0.57	0.71	**0.90**	0.42	0.65	0.15	0.11
LFR, $\langle k \rangle$=10		0.46$^-$	0.47	0.36	0.12	0.51	0.36	**0.79**	**0.79**	0.71	0.61	0.36	0.63	0.53	0.78	0.27	0.31	0.15	0.13
LFR, o_m=4		0.62$^-$	0.63	0.62	0.44	0.60	0.52	0.71	0.71	0.71	0.62	0.40	0.67	0.77	**0.80**	0.43	0.66	0.40	0.29
LFR, sizes 50–250		0.42$^-$	0.43	0.44	0.16	0.46	0.16	0.87	0.85	0.79	0.75	0.38	0.79	0.62	**0.91**	0.33	0.55	0.14	0.08
LFR, C_{LFR}=0.3		0.39$^-$	0.40	0.39	0.24	0.31	0.42	**0.62**	0.61	0.58	0.52	0.48	0.48	0.59	0.59	0.27	0.36	0.23	0.22
LFR, C_{LFR}=0.5		0.30$^-$	0.30	0.29	0.24	0.25	0.28	**0.38**	**0.37**	0.35	0.26	0.35	0.30	0.37	0.36	0.22	0.27	0.22	0.23

TABLE S9. ONMI (McDaid *et al.*, max normalisation) of the oracle covers of Table S8; superscripts on HC as in Table S6, for ONMI. Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods					
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	OSLOM	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.52$^-$	0.51	0.41	0.47	0.49	0.41	**0.61**	**0.61**	0.36	0.38	0.00	0.35	0.15	0.49	0.44	0.46	0.45	0.47
	0.05	**0.47**$^+$	0.46	0.39	0.20	0.42	0.39	0.46	0.45	0.18	0.22	0.00	0.23	0.12	0.36	0.28	0.28	0.16	0.19
	0.10	**0.41**$^+$	0.40	0.37	0.12	0.33	0.35	0.27	0.27	0.11	0.11	0.00	0.14	0.10	0.26	0.23	0.27	0.09	0.10
uniform, k=16	0.00	0.71$^-$	0.71	0.67	0.65	0.51	0.59	**0.83**	0.82	0.31	0.62	0.01	0.41	0.09	0.63	0.53	0.56	0.59	0.54
	0.05	0.66$^-$	0.65	0.62	0.29	0.32	0.46	**0.71**	**0.70**	0.16	0.42	0.01	0.25	0.09	0.55	0.29	0.52	0.22	0.26
	0.10	**0.56**$^=$	**0.55**	0.52	0.19	0.18	0.45	**0.55**	0.54	0.11	0.26	0.01	0.14	0.08	0.45	0.22	0.46	0.13	0.17
equal k, unequal s, k=8	0.00	0.49$^-$	0.48	0.38	0.44	0.46	0.38	**0.55**	**0.54**	0.24	0.28	0.00	0.25	0.08	0.40	0.41	0.42	0.41	0.43
	0.05	**0.44**$^+$	0.43	0.36	0.15	0.38	0.35	0.31	0.31	0.10	0.13	0.00	0.14	0.06	0.25	0.24	0.23	0.11	0.14
	0.10	**0.36**$^+$	0.36	0.32	0.08	0.30	0.30	0.17	0.17	0.06	0.06	0.00	0.08	0.05	0.16	0.20	0.21	0.05	0.08
equal k, unequal s, k=16	0.00	0.74$^-$	0.73	0.69	0.66	0.53	0.62	**0.85**	**0.85**	0.52	0.71	0.07	0.43	0.26	0.71	0.57	0.60	0.61	0.55
	0.05	0.69$^-$	0.68	0.65	0.36	0.37	0.50	**0.74**	0.73	0.42	0.56	0.06	0.33	0.26	0.65	0.39	0.57	0.29	0.31
	0.10	**0.61**$^=$	**0.61**	0.57	0.24	0.23	0.48	**0.61**	0.60	0.36	0.44	0.05	0.27	0.24	0.59	0.30	0.52	0.18	0.21
equal s, unequal k	0.00	0.67$^-$	0.67	0.61	0.62	0.38	0.42	**0.76**	0.70	0.38	0.56	0.05	0.49	0.14	0.45	0.40	0.46	0.58	0.52
	0.05	0.57$^-$	0.57	0.53	0.31	0.21	0.40	**0.64**	0.58	0.27	0.47	0.04	0.36	0.13	0.40	0.21	0.39	0.25	0.28
	0.10	0.45$^-$	0.44	0.42	0.24	0.12	0.36	**0.52**	0.51	0.23	0.40	0.04	0.28	0.11	0.35	0.15	0.27	0.19	0.22
two worlds	0.00	0.62$^-$	0.60	0.54	0.56	0.34	0.32	0.54	**0.70**	0.26	0.38	0.01	0.23	0.09	0.25	0.34	0.38	0.53	0.48
	0.05	0.42$^-$	0.43	0.40	0.22	0.15	0.28	0.35	**0.57**	0.14	0.25	0.01	0.18	0.06	0.20	0.15	0.28	0.19	0.21
	0.10	0.28$^-$	0.28	0.27	0.17	0.08	0.24	0.24	**0.43**	0.09	0.18	0.01	0.13	0.04	0.14	0.11	0.16	0.14	0.16
three big, many small	0.00	0.62$^-$	0.61	0.55	0.57	0.30	0.33	0.32	**0.69**	0.15	0.19	0.00	0.18	0.06	0.16	0.29	0.36	0.54	0.48
	0.05	0.40$^-$	0.40	0.38	0.19	0.11	0.29	0.18	**0.52**	0.07	0.14	0.00	0.10	0.04	0.13	0.13	0.21	0.17	0.20
	0.10	0.26$^=$	**0.26**	0.25	0.15	0.06	0.24	0.11	**0.27**	0.04	0.10	0.00	0.07	0.02	0.08	0.09	0.12	0.12	0.15
gradient	0.00	**0.51**$^+$	0.50	0.43	0.46	0.42	0.40	0.29	0.34	0.09	0.05	0.00	0.13	0.01	0.14	0.37	0.32	0.45	0.44
	0.05	**0.38**$^+$	**0.37**	0.33	0.12	0.18	0.20	0.09	0.11	0.02	0.03	0.00	0.04	0.01	0.08	0.16	0.25	0.08	0.11
	0.10	**0.23**$^+$	**0.23**	0.20	0.08	0.08	0.19	0.03	0.05	0.01	0.02	0.00	0.02	0.01	0.05	0.10	0.17	0.05	0.07
uniform k=8, α=1	0.05	**0.47**$^+$	0.46	0.39	0.22	0.42	0.39	0.46	0.46	0.18	0.21	0.00	0.23	0.13	0.36	0.28	0.28	0.16	0.20
uniform k=8, α=2	0.05	**0.48**$^+$	0.47	0.39	0.25	0.41	0.38	0.46	0.46	0.18	0.20	0.00	0.23	0.12	0.36	0.28	0.28	0.19	0.23
two worlds, α=1	0.05	0.51$^-$	0.51	0.44	0.31	0.17	0.26	0.39	**0.58**	0.18	0.23	0.01	0.17	0.08	0.22	0.13	0.22	0.24	0.28
two worlds, α=2	0.05	**0.60**$^+$	**0.59**	0.47	0.41	0.18	0.25	0.44	0.56	0.21	0.22	0.01	0.17	0.09	0.24	0.11	0.20	0.32	0.35
LFR, o_n=0		0.80$^-$	0.80	0.77	0.60	0.38	0.54	**1.00**	**1.00**	**1.00**	0.98	0.65	0.99	0.92	**1.00**	0.37	0.69	0.52	0.37
LFR, o_n=0.1		0.70$^-$	0.71	0.68	0.28	0.24	0.51	0.87	0.87	0.87	0.79	0.49	0.84	0.85	**0.96**	0.28	0.64	0.29	0.22
LFR, o_n=0.3		0.49$^-$	0.51	0.49	0.12	0.12	0.45	0.69	0.69	0.67	0.55	0.28	0.58	0.70	**0.89**	0.21	0.53	0.14	0.10
LFR, $\langle k \rangle$=10		0.37$^-$	0.37	0.26	0.08	0.29	0.25	**0.77**	**0.77**	0.68	0.57	0.36	0.62	0.37	**0.76**	0.22	0.23	0.11	0.09
LFR, o_m=4		0.59$^-$	0.60	0.59	0.41	0.20	0.50	0.69	0.69	0.69	0.60	0.37	0.63	0.73	**0.78**	0.35	0.59	0.39	0.30
LFR, sizes 50–250		0.34$^-$	0.35	0.30	0.11	0.12	0.10	0.84	0.83	0.77	0.72	0.37	0.77	0.63	**0.90**	0.15	0.42	0.10	0.06
LFR, C_{LFR}=0.3		0.31$^-$	0.32	0.31	0.17	0.08	0.31	**0.57**	0.56	0.54	0.48	0.45	0.39	0.51	0.54	0.13	0.20	0.17	0.16
LFR, C_{LFR}=0.5		0.19$^-$	0.19	0.18	0.13	0.04	0.13	**0.30**	0.28	0.25	0.18	0.26	0.17	0.22	0.26	0.08	0.08	0.12	0.13

TABLE S10. ONMI (McDaid *et al.*, max normalisation) of the label-free covers of Table S6; superscripts on HC as in Table S6, for ONMI. Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods					
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	OSLOM	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.52$^-$	0.51	0.41	0.46	0.49	0.41	**0.57**	**0.57**	0.36	0.38	0.00	0.31	0.07	0.47	0.44	0.46	0.45	0.47
	0.05	**0.47**$^+$	0.46	0.39	0.20	0.43	0.39	0.36	0.35	0.17	0.22	0.00	0.18	0.06	0.35	0.28	0.29	0.17	0.19
	0.10	**0.41**$^+$	0.41	0.36	0.12	0.36	0.35	0.20	0.19	0.09	0.11	0.00	0.10	0.04	0.25	0.22	0.27	0.09	0.11
uniform, k=16	0.00	0.72$^-$	0.71	0.67	0.65	0.52	0.60	**0.82**	0.82	0.31	0.62	0.01	0.41	0.05	0.61	0.53	0.56	0.59	0.54
	0.05	0.66$^-$	0.65	0.62	0.30	0.37	0.46	**0.70**	**0.69**	0.17	0.42	0.01	0.24	0.05	0.52	0.33	0.53	0.23	0.26
	0.10	**0.56**$^+$	**0.55**	0.52	0.19	0.24	0.45	0.53	0.52	0.11	0.26	0.01	0.14	0.05	0.41	0.25	0.47	0.14	0.17
equal k, unequal s, k=8	0.00	0.49$^-$	0.48	0.38	0.44	0.46	0.38	**0.52**	0.51	0.24	0.28	0.00	0.25	0.03	0.39	0.40	0.42	0.42	0.44
	0.05	**0.44**$^+$	0.43	0.36	0.16	0.40	0.35	0.27	0.27	0.10	0.13	0.00	0.13	0.03	0.24	0.22	0.23	0.12	0.14
	0.10	**0.37**$^+$	0.35	0.32	0.09	0.33	0.30	0.13	0.13	0.05	0.06	0.00	0.06	0.02	0.15	0.16	0.21	0.06	0.08
equal k, unequal s, k=16	0.00	0.74$^-$	0.73	0.69	0.67	0.54	0.63	**0.82**	**0.82**	0.52	0.71	0.07	0.41	0.23	0.70	0.56	0.60	0.61	0.55
	0.05	**0.68**$^+$	**0.68**	0.64	0.36	0.40	0.50	0.67	0.67	0.42	0.56	0.06	0.30	0.23	0.64	0.43	0.57	0.31	0.30
	0.10	**0.60**$^+$	**0.60**	0.57	0.23	0.29	0.48	0.51	0.50	0.36	0.44	0.05	0.23	0.21	0.58	0.37	0.52	0.19	0.21
equal s, unequal k	0.00	0.67$^-$	0.67	0.61	0.63	0.36	0.40	**0.76**	0.71	0.39	0.56	0.05	0.50	0.14	0.46	0.46	0.46	0.59	0.52
	0.05	0.55$^-$	0.54	0.51	0.31	0.23	0.40	**0.61**	0.58	0.28	0.47	0.04	0.36	0.13	0.39	0.31	0.40	0.26	0.28
	0.10	0.44$^-$	0.45	0.42	0.24	0.13	0.37	0.49	**0.50**	0.23	0.40	0.04	0.27	0.12	0.35	0.25	0.35	0.19	0.22
two worlds	0.00	0.62$^-$	0.61	0.54	0.56	0.29	0.31	0.48	**0.67**	0.26	0.38	0.01	0.23	0.05	0.16	0.39	0.39	0.53	0.48
	0.05	**0.45**$^=$	**0.45**	0.41	0.22	0.17	0.30	0.29	0.45	0.14	0.25	0.01	0.18	0.04	0.09	0.26	0.34	0.19	0.21
	0.10	**0.28**$^+$	**0.29**	0.27	0.17	0.09	0.25	0.19	0.27	0.09	0.18	0.01	0.13	0.03	0.05	0.20	0.26	0.14	0.16
three big, many small	0.00	**0.63**$^+$	0.62	0.55	0.57	0.25	0.32	0.32	0.37	0.14	0.19	0.00	0.15	0.03	0.11	0.33	0.38	0.54	0.48
	0.05	0.39$^+$	**0.40**	0.37	0.17	0.13	0.28	0.19	0.27	0.08	0.14	0.00	0.09	0.02	0.06	0.13	0.26	0.15	0.17
	0.10	**0.25**$^+$	**0.24**	0.23	0.12	0.06	0.22	0.11	0.18	0.04	0.10	0.00	0.06	0.01	0.03	0.09	0.15	0.11	0.14
gradient	0.00	0.27$^-$	0.24	0.25	0.23	**0.40**	0.38	0.20	0.16	0.05	0.05	0.00	0.10	0.01	0.13	0.35	0.29	0.31	0.26
	0.05	0.15$^-$	0.14	0.11	0.07	0.19	0.18	0.08	0.09	0.02	0.03	0.00	0.03	0.01	0.07	0.09	**0.23**	0.07	0.08
	0.10	0.07$^-$	0.07	0.06	0.03	0.09	**0.16**	0.03	0.04	0.01	0.02	0.00	0.02	0.01	0.04	0.05	0.15	0.04	0.05
uniform k=8, α=1	0.05	**0.47**$^+$	0.46	0.39	0.22	0.43	0.39	0.36	0.36	0.18	0.21	0.00	0.19	0.05	0.35	0.28	0.28	0.18	0.21
uniform k=8, α=2	0.05	**0.48**$^+$	0.47	0.39	0.25	0.43	0.38	0.36	0.36	0.17	0.20	0.00	0.18	0.05	0.35	0.28	0.29	0.21	0.23
two worlds, α=1	0.05	**0.52**$^+$	**0.51**	0.43	0.31	0.19	0.28	0.33	0.44	0.17	0.23	0.01	0.16	0.05	0.11	0.24	0.30	0.24	0.28
two worlds, α=2	0.05	**0.59**$^+$	**0.59**	0.46	0.43	0.19	0.28	0.37	0.44	0.19	0.22	0.01	0.16	0.06	0.14	0.22	0.27	0.33	0.36
LFR, o_n=0		0.80$^-$	0.80	0.77	0.57	0.38	0.54	0.97	0.97	**1.00**	0.98	0.65	0.99	0.92	**0.99**	0.47	0.70	0.49	0.36
LFR, o_n=0.1		0.66$^-$	0.69	0.67	0.26	0.24	0.51	0.68	0.68	0.86	0.79	0.49	0.84	0.79	**0.92**	0.41	0.66	0.28	0.22
LFR, o_n=0.3		0.44$^-$	0.48	0.44	0.10	0.12	0.45	0.35	0.35	0.64	0.55	0.28	0.38	0.59	**0.83**	0.38	0.57	0.13	0.10
LFR, $\langle k \rangle$=10		0.36$^-$	0.37	0.26	0.07	0.30	0.25	0.58	0.60	0.67	0.57	0.36	0.62	0.40	**0.74**	0.23	0.19	0.10	0.09
LFR, o_m=4		0.58$^-$	0.60	0.58	0.38	0.19	0.50	0.47	0.47	0.68	0.60	0.37	0.63	0.64	**0.71**	0.42	0.61	0.37	0.31
LFR, sizes 50–250		0.35$^-$	0.34	0.30	0.10	0.13	0.10	0.84	0.82	0.77	0.72	0.37	0.77	0.59	**0.88**	0.17	0.42	0.10	0.06
LFR, C_{LFR}=0.3		0.28$^-$	0.28	0.28	0.15	0.08	0.29	0.35	0.35	0.47	0.48	0.45	0.22	**0.50**	0.46	0.19	0.23	0.16	0.16
LFR, C_{LFR}=0.5		0.16$^-$	0.16	0.15	0.11	0.04	0.11	0.12	0.12	0.17	0.18	**0.26**	0.11	0.24	0.23	0.10	0.10	0.10	0.12

TABLE S11. Per-community recovery of the true communities no larger than the median size, oracle covers, on the unequal-ring families (equal s with unequal k has no community larger than the median). Numbers of rings no larger than the median size: 20, 20, 20, 20, 16, 5 (families in order). Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods					
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	OSLOM	DEMON	ANGEL	SCAN	SCAN-J
equal k, unequal s, k=8	0.00	0.62	0.62	0.52	0.57	0.62	0.52	**0.68**	**0.67**	0.47	0.49	0.15	0.32	0.39	0.55	0.57	0.55	0.55	0.55
	0.05	**0.58**	**0.58**	0.51	0.35	0.57	0.50	0.53	0.53	0.34	0.35	0.14	0.28	0.37	0.46	0.48	0.42	0.32	0.33
	0.10	**0.52**	0.51	0.47	0.28	**0.52**	0.45	0.41	0.41	0.29	0.28	0.13	0.24	0.35	0.38	0.44	0.41	0.26	0.26
equal k, unequal s, k=16	0.00	0.82	0.82	0.80	0.76	0.82	0.75	**0.91**	0.90	0.78	0.83	0.23	0.53	0.59	0.84	0.77	0.74	0.71	0.65
	0.05	0.78	0.78	0.76	0.54	0.74	0.66	**0.83**	0.81	0.74	0.73	0.19	0.46	0.57	0.80	0.72	0.71	0.48	0.48
	0.10	0.72	0.72	0.69	0.44	0.65	0.63	0.73	0.72	0.66	0.66	0.17	0.43	0.54	**0.77**	0.67	0.67	0.38	0.39
equal s, unequal k	0.00	0.77	0.77	0.73	0.72	0.75	0.57	**0.80**	0.78	0.50	0.65	0.17	0.54	0.33	0.52	0.60	0.61	0.68	0.62
	0.05	0.68	0.69	0.66	0.47	0.61	0.55	**0.70**	0.68	0.40	0.56	0.15	0.44	0.30	0.47	0.48	0.55	0.41	0.42
	0.10	0.58	0.58	0.56	0.40	0.52	0.51	**0.61**	**0.61**	0.35	0.51	0.14	0.38	0.28	0.43	0.42	0.47	0.33	0.35
two worlds	0.00	0.61	0.60	0.52	0.56	0.56	0.27	0.47	**0.68**	0.39	0.38	0.13	0.18	0.30	0.55	0.45	0.41	0.54	0.50
	0.05	0.38	0.39	0.35	0.20	0.30	0.20	0.26	**0.54**	0.30	0.28	0.11	0.15	0.26	0.44	0.31	0.33	0.20	0.22
	0.10	0.21	0.21	0.19	0.16	0.21	0.14	0.15	**0.38**	0.25	0.23	0.09	0.13	0.23	0.33	0.27	0.24	0.15	0.18
three big, many small	0.00	0.60	0.59	0.53	0.57	0.56	0.26	0.22	**0.69**	0.35	0.32	0.12	0.10	0.30	0.46	0.42	0.38	0.53	0.50
	0.05	0.33	0.34	0.31	0.19	0.27	0.20	0.11	**0.53**	0.29	0.25	0.11	0.14	0.26	0.37	0.30	0.28	0.19	0.22
	0.10	0.17	0.17	0.16	0.16	0.19	0.14	0.14	**0.37**	0.24	0.21	0.09	0.13	0.22	0.26	0.27	0.21	0.15	0.17
gradient	0.00	**0.59**	**0.59**	0.50	0.56	0.57	0.49	0.32	0.52	0.35	0.37	0.13	0.19	0.36	0.33	0.49	0.42	0.53	0.52
	0.05	**0.48**	**0.48**	0.42	0.29	0.41	0.32	0.27	0.29	0.28	0.29	0.12	0.18	0.34	0.26	0.41	0.37	0.24	0.27
	0.10	0.35	0.35	0.31	0.22	0.32	0.30	0.22	0.20	0.25	0.24	0.11	0.16	0.30	0.20	**0.36**	0.31	0.19	0.20

TABLE S12. Per-community recovery of the true communities larger than the median size, oracle covers, on the unequal-ring families (equal s with unequal k has no community larger than the median). Numbers of rings larger than the median size: 10, 10, 5, 3, 5 (families in order). Bold: best in row, within 0.005.

		HC and ablations				relatives		partitions						overlapping methods					
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	OSLOM	DEMON	ANGEL	SCAN	SCAN-J
equal k, unequal s, k=8	0.00	0.56	0.55	0.44	0.50	0.55	0.44	**0.57**	**0.57**	0.28	0.32	0.11	0.32	0.22	0.46	0.49	0.47	0.48	0.49
	0.05	**0.50**	0.49	0.43	0.25	0.49	0.41	0.32	0.31	0.19	0.20	0.10	0.21	0.21	0.31	0.40	0.31	0.22	0.24
	0.10	**0.43**	0.42	0.38	0.20	0.42	0.36	0.24	0.23	0.17	0.17	0.10	0.16	0.20	0.24	0.36	0.29	0.17	0.18
equal k, unequal s, k=16	0.00	0.79	0.79	0.76	0.72	0.79	0.70	**0.85**	**0.86**	0.43	0.69	0.15	0.45	0.33	0.71	0.71	0.67	0.67	0.61
	0.05	0.74	0.74	0.71	0.46	0.69	0.58	**0.76**	**0.76**	0.30	0.53	0.14	0.36	0.32	0.64	0.63	0.64	0.39	0.40
	0.10	**0.66**	**0.66**	0.64	0.35	0.58	0.56	0.62	0.62	0.27	0.40	0.13	0.30	0.31	0.57	0.56	0.59	0.30	0.31
two worlds	0.00	**0.88**	**0.88**	0.87	0.81	0.87	0.77	0.73	0.87	0.38	0.62	0.17	0.48	0.27	0.24	0.65	0.71	0.78	0.70
	0.05	**0.85**	**0.85**	0.84	0.67	0.81	0.76	0.68	0.83	0.26	0.51	0.16	0.41	0.25	0.24	0.55	0.64	0.58	0.56
	0.10	**0.81**	**0.81**	0.80	0.58	0.72	0.74	0.60	0.79	0.24	0.42	0.16	0.37	0.24	0.24	0.51	0.55	0.50	0.51
three big, many small	0.00	**0.87**	**0.87**	0.86	0.80	0.86	0.78	0.58	0.82	0.23	0.28	0.18	0.44	0.17	0.16	0.63	0.70	0.78	0.71
	0.05	**0.84**	**0.84**	0.83	0.60	0.77	0.76	0.47	0.72	0.15	0.26	0.19	0.30	0.17	0.16	0.55	0.58	0.52	0.54
	0.10	0.77	**0.77**	0.74	0.53	0.68	0.72	0.31	0.47	0.13	0.23	0.20	0.26	0.17	0.16	0.47	0.51	0.44	0.48
gradient	0.00	**0.60**	0.59	0.52	0.54	0.59	0.50	0.40	0.44	0.17	0.15	0.12	0.27	0.11	0.29	0.50	0.44	0.51	0.49
	0.05	**0.49**	**0.49**	0.43	0.25	0.42	0.32	0.20	0.26	0.10	0.13	0.13	0.20	0.11	0.21	0.37	0.38	0.20	0.21
	0.10	**0.37**	**0.36**	0.33	0.22	0.33	0.31	0.15	0.22	0.09	0.12	0.14	0.17	0.10	0.17	0.31	0.31	0.18	0.17

TABLE S13. Coverage (fraction of nodes in at least one community) of the label-free covers.

		HC and ablations				relatives		partitions						overlapping methods					
Family	o_n	HC	e-c.	tri.	2-test	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	OSLOM	DEMON	ANGEL	SCAN	SCAN-J
uniform, k=8	0.00	0.96	0.97	0.79	0.82	0.97	0.79	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.71	0.71	0.80	0.74
	0.05	0.96	0.97	0.80	0.77	0.97	0.80	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.72	0.71	0.73	0.68
	0.10	0.96	0.96	0.81	0.75	0.97	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.73	0.72	0.68	0.63
uniform, k=16	0.00	0.95	0.95	0.96	0.86	0.97	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.93	0.91	0.84	0.79
	0.05	0.94	0.94	0.95	0.78	0.97	0.93	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.93	0.92	0.75	0.71
	0.10	0.92	0.92	0.94	0.75	0.97	0.92	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.94	0.92	0.69	0.67
equal k, unequal s, k=8	0.00	0.96	0.97	0.79	0.82	0.97	0.79	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.71	0.71	0.80	0.74
	0.05	0.96	0.96	0.80	0.76	0.97	0.80	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.72	0.71	0.73	0.68
	0.10	0.96	0.96	0.81	0.74	0.97	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.73	0.72	0.68	0.63
equal k, unequal s, k=16	0.00	0.95	0.95	0.96	0.86	0.97	0.97	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.93	0.91	0.84	0.80
	0.05	0.94	0.94	0.95	0.80	0.97	0.95	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.93	0.92	0.76	0.72
	0.10	0.92	0.92	0.95	0.75	0.97	0.91	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.94	0.92	0.70	0.67
equal s, unequal k	0.00	0.93	0.93	0.94	0.86	0.96	0.85	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.93	0.91	0.83	0.83
	0.05	0.92	0.92	0.93	0.77	0.95	0.84	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.94	0.92	0.71	0.72
	0.10	0.88	0.90	0.90	0.71	0.94	0.82	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.95	0.93	0.64	0.67
two worlds	0.00	0.93	0.93	0.90	0.85	0.92	0.75	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.90	0.87	0.80	0.79
	0.05	0.89	0.89	0.87	0.67	0.92	0.74	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.91	0.88	0.63	0.66
	0.10	0.81	0.82	0.81	0.62	0.91	0.70	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.93	0.89	0.57	0.63
three big, many small	0.00	0.92	0.92	0.90	0.85	0.94	0.79	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.92	0.90	0.81	0.78
	0.05	0.84	0.86	0.85	0.64	0.93	0.75	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.94	0.91	0.60	0.63
	0.10	0.77	0.78	0.77	0.60	0.91	0.68	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.95	0.91	0.55	0.64
gradient	0.00	0.93	0.93	0.87	0.82	0.95	0.88	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.83	0.82	0.80	0.74
	0.05	0.91	0.92	0.86	0.76	0.95	0.86	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.84	0.82	0.72	0.70
	0.10	0.90	0.91	0.86	0.72	0.95	0.84	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.85	0.83	0.71	0.66
uniform k=8, α=1	0.05	0.97	0.97	0.80	0.78	0.98	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.72	0.71	0.75	0.68
uniform k=8, α=2	0.05	0.97	0.97	0.81	0.80	0.98	0.81	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.73	0.72	0.78	0.71
two worlds, α=1	0.05	0.91	0.92	0.86	0.82	0.94	0.71	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.89	0.87	0.75	0.74
two worlds, α=2	0.05	0.93	0.93	0.86	0.88	0.95	0.71	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.89	0.87	0.80	0.79
LFR, o_n=0		0.97	0.97	0.97	0.87	0.99	0.79	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.97	0.86	0.65
LFR, o_n=0.1		0.96	0.96	0.95	0.75	0.99	0.77	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.96	0.70	0.61
LFR, o_n=0.3		0.91	0.91	0.91	0.58	0.98	0.73	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.95	0.55	0.52
LFR, $\langle k \rangle$=10		0.90	0.91	0.69	0.58	0.95	0.70	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00	0.59	0.57	0.53	0.39
LFR, o_m=4		0.95	0.95	0.94	0.82	0.99	0.79	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.96	0.81	0.68
LFR, sizes 50–250		0.86	0.87	0.86	0.54	0.94	0.60	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.91	0.86	0.47	0.45
LFR, C_{LFR}=0.3		0.88	0.88	0.88	0.71	0.99	0.84	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.67	0.65
LFR, C_{LFR}=0.5		0.93	0.93	0.93	0.87	1.00	0.92	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.84	0.84

TABLE S14. The paired intervals behind Table I: Ω of HyperCommon minus that of the best (first line of each cell) and second-best other method, with 95% bootstrap intervals over the ten test seeds, ablations excluded.

Family	o_n	oracle	label-free	frozen	native
uniform, k=8	0.00	Leiden −0.054 [−0.057, −0.051]	Leiden −0.001 [−0.009, +0.007]	Leiden −0.080 [−0.084, −0.078]	Leiden −0.003 [−0.010, +0.003]
		L-CPM −0.050 [−0.053, −0.047]	L-CPM +0.002 [−0.004, +0.007]	L-CPM −0.080 [−0.082, −0.078]	L-CPM +0.004 [−0.000, +0.009]
	0.05	Link +0.005 [+0.002, +0.007]	Link +0.024 [+0.018, +0.030]	Leiden −0.026 [−0.033, −0.019]	CPM +0.063 [+0.058, +0.068]
		Leiden +0.020 [+0.016, +0.026]	CPM +0.063 [+0.058, +0.068]	L-CPM −0.020 [−0.026, −0.014]	Leiden +0.103 [+0.097, +0.109]
	0.10	Link +0.009 [+0.004, +0.013]	Link +0.049 [+0.039, +0.057]	CPM +0.005 [−0.001, +0.010]	CPM +0.063 [+0.054, +0.073]
		CPM +0.079 [+0.071, +0.086]	CPM +0.063 [+0.054, +0.073]	Link +0.008 [+0.001, +0.014]	ANGEL +0.095 [+0.087, +0.102]
uniform, k=16	0.00	Leiden −0.078 [−0.082, −0.074]	Leiden −0.065 [−0.070, −0.060]	Leiden −0.101 [−0.106, −0.096]	Leiden −0.065 [−0.068, −0.061]
		L-CPM −0.073 [−0.076, −0.069]	L-CPM −0.058 [−0.062, −0.054]	L-CPM −0.097 [−0.103, −0.092]	L-CPM −0.059 [−0.063, −0.056]
	0.05	Leiden −0.044 [−0.048, −0.039]	Leiden −0.035 [−0.040, −0.029]	Leiden −0.193 [−0.207, −0.178]	Leiden −0.037 [−0.043, −0.030]
		L-CPM −0.040 [−0.045, −0.035]	L-CPM −0.029 [−0.037, −0.022]	L-CPM −0.186 [−0.200, −0.169]	L-CPM −0.029 [−0.036, −0.021]
	0.10	Leiden −0.008 [−0.017, +0.000]	Leiden +0.002 [−0.010, +0.011]	Leiden −0.316 [−0.328, −0.305]	Leiden +0.003 [−0.007, +0.010]
		L-CPM +0.001 [−0.009, +0.009]	L-CPM +0.016 [+0.007, +0.025]	L-CPM −0.308 [−0.320, −0.297]	L-CPM +0.012 [+0.002, +0.020]
equal k, unequal s, k=8	0.00	Leiden −0.028 [−0.031, −0.024]	Leiden +0.012 [+0.008, +0.016]	Leiden −0.044 [−0.058, −0.033]	Leiden +0.010 [+0.008, +0.013]
		L-CPM −0.025 [−0.028, −0.021]	Link +0.013 [+0.008, +0.018]	L-CPM −0.041 [−0.052, −0.031]	L-CPM +0.013 [+0.010, +0.017]
	0.05	Link +0.008 [+0.003, +0.012]	Link +0.029 [+0.023, +0.035]	Link +0.000 [−0.004, +0.006]	CPM +0.093 [+0.085, +0.100]
		CPM +0.095 [+0.088, +0.101]	CPM +0.093 [+0.085, +0.100]	CPM +0.033 [+0.029, +0.038]	DEMON +0.144 [+0.139, +0.150]
	0.10	Link +0.002 [−0.003, +0.008]	Link +0.038 [+0.029, +0.048]	CPM +0.000 [−0.010, +0.012]	CPM +0.084 [+0.073, +0.095]
		CPM +0.084 [+0.073, +0.097]	CPM +0.084 [+0.073, +0.096]	Link +0.003 [−0.005, +0.012]	DEMON +0.118 [+0.111, +0.126]
equal k, unequal s, k=16	0.00	Leiden −0.074 [−0.078, −0.071]	Leiden −0.045 [−0.048, −0.041]	Leiden −0.101 [−0.104, −0.096]	Leiden −0.044 [−0.048, −0.040]
		L-CPM −0.072 [−0.074, −0.068]	L-CPM −0.040 [−0.045, −0.035]	L-CPM −0.093 [−0.096, −0.091]	L-CPM −0.040 [−0.044, −0.036]
	0.05	Leiden −0.039 [−0.042, −0.036]	Leiden +0.004 [−0.004, +0.011]	Leiden −0.173 [−0.185, −0.162]	Leiden +0.009 [+0.002, +0.017]
		L-CPM −0.030 [−0.034, −0.026]	L-CPM +0.007 [−0.001, +0.014]	L-CPM −0.162 [−0.174, −0.147]	L-CPM +0.015 [+0.007, +0.022]
	0.10	Leiden +0.010 [+0.001, +0.018]	OSLOM +0.015 [−0.001, +0.030]	Leiden −0.306 [−0.319, −0.291]	ANGEL +0.039 [+0.027, +0.049]
		L-CPM +0.016 [+0.008, +0.022]	ANGEL +0.039 [+0.028, +0.049]	L-CPM −0.303 [−0.316, −0.289]	Leiden +0.059 [+0.048, +0.069]
equal s, unequal k	0.00	Leiden −0.028 [−0.031, −0.025]	L-CPM −0.005 [−0.012, +0.002]	Leiden −0.070 [−0.073, −0.067]	Leiden +0.000 [−0.006, +0.006]
		L-CPM −0.027 [−0.030, −0.024]	Leiden −0.002 [−0.009, +0.004]	L-CPM −0.036 [−0.039, −0.033]	L-CPM +0.033 [+0.028, +0.037]
	0.05	L-CPM −0.011 [−0.016, −0.006]	L-CPM −0.006 [−0.012, +0.001]	Leiden −0.183 [−0.202, −0.165]	L-CPM +0.025 [+0.017, +0.033]
		Leiden −0.006 [−0.014, +0.001]	Leiden +0.021 [+0.013, +0.031]	L-CPM −0.179 [−0.195, −0.165]	Leiden +0.027 [+0.019, +0.034]
	0.10	L-CPM −0.029 [−0.034, −0.023]	L-CPM −0.017 [−0.024, −0.009]	L-CPM −0.328 [−0.337, −0.319]	L-CPM +0.028 [+0.020, +0.037]
		Leiden −0.006 [−0.013, +0.001]	Leiden +0.031 [+0.024, +0.039]	Leiden −0.316 [−0.325, −0.306]	Leiden +0.031 [+0.024, +0.039]
two worlds	0.00	L-CPM −0.004 [−0.007, −0.000]	L-CPM +0.035 [+0.027, +0.042]	L-CPM −0.051 [−0.056, −0.047]	L-CPM +0.007 [+0.001, +0.011]
		Link +0.019 [+0.013, +0.027]	Link +0.042 [+0.035, +0.050]	Link −0.001 [−0.004, +0.003]	SCAN +0.104 [+0.090, +0.119]
	0.05	L-CPM −0.007 [−0.016, −0.000]	CPM +0.100 [+0.079, +0.116]	L-CPM −0.249 [−0.268, −0.230]	L-CPM +0.001 [−0.013, +0.013]
		Link +0.065 [+0.054, +0.077]	Link +0.103 [+0.085, +0.121]	CPM −0.121 [−0.139, −0.102]	CPM +0.100 [+0.078, +0.117]
	0.10	L-CPM −0.004 [−0.012, +0.004]	CPM +0.063 [+0.052, +0.074]	L-CPM −0.383 [−0.397, −0.369]	L-CPM +0.009 [−0.001, +0.019]
		CPM +0.061 [+0.051, +0.071]	Link +0.184 [+0.164, +0.202]	CPM −0.189 [−0.205, −0.173]	CPM +0.063 [+0.052, +0.074]
three big, many small	0.00	Link +0.009 [+0.005, +0.017]	Link +0.025 [+0.019, +0.034]	Link +0.026 [+0.014, +0.036]	SCAN +0.092 [+0.082, +0.103]
		L-CPM +0.032 [+0.025, +0.039]	SCAN +0.092 [+0.081, +0.103]	CPM +0.100 [+0.089, +0.112]	CPM +0.112 [+0.103, +0.120]
	0.05	Link +0.075 [+0.059, +0.090]	CPM +0.102 [+0.084, +0.121]	CPM −0.159 [−0.177, −0.140]	CPM +0.102 [+0.083, +0.122]
		CPM +0.084 [+0.073, +0.097]	Link +0.103 [+0.076, +0.125]	L-CPM −0.108 [−0.131, −0.083]	ANGEL +0.264 [+0.249, +0.277]
	0.10	CPM +0.043 [+0.026, +0.059]	CPM +0.059 [+0.033, +0.086]	CPM −0.230 [−0.243, −0.217]	CPM +0.059 [+0.033, +0.086]
		Link +0.098 [+0.077, +0.118]	Link +0.122 [+0.087, +0.152]	ANGEL −0.117 [−0.129, −0.106]	ANGEL +0.239 [+0.222, +0.258]
gradient	0.00	Link −0.012 [−0.022, −0.003]	Link −0.186 [−0.210, −0.165]	Link −0.002 [−0.018, +0.014]	CPM −0.139 [−0.165, −0.116]
		CPM +0.078 [+0.064, +0.095]	CPM −0.139 [−0.166, −0.115]	CPM +0.024 [+0.009, +0.041]	DEMON −0.080 [−0.103, −0.058]
	0.05	Link +0.033 [+0.016, +0.049]	Link −0.162 [−0.177, −0.147]	ANGEL −0.022 [−0.046, +0.002]	ANGEL −0.107 [−0.125, −0.091]
		ANGEL +0.103 [+0.085, +0.119]	ANGEL −0.107 [−0.126, −0.091]	DEMON −0.003 [−0.019, +0.012]	CPM −0.087 [−0.098, −0.073]
	0.10	Link +0.028 [+0.016, +0.039]	Link −0.152 [−0.177, −0.127]	DEMON −0.136 [−0.152, −0.121]	CPM −0.116 [−0.138, −0.088]
		CPM +0.072 [+0.060, +0.084]	CPM −0.116 [−0.137, −0.090]	ANGEL −0.119 [−0.133, −0.105]	ANGEL −0.095 [−0.122, −0.068]
uniform k=8, α=1	0.05	Link +0.008 [+0.005, +0.011]	Link +0.035 [+0.027, +0.043]	Leiden −0.027 [−0.033, −0.022]	CPM +0.079 [+0.074, +0.084]
		Leiden +0.019 [+0.014, +0.023]	CPM +0.079 [+0.074, +0.084]	L-CPM −0.021 [−0.027, −0.016]	Leiden +0.116 [+0.110, +0.123]
uniform k=8, α=2	0.05	Link +0.010 [+0.007, +0.013]	Link +0.024 [+0.018, +0.031]	Leiden −0.021 [−0.027, −0.016]	CPM +0.081 [+0.076, +0.087]
		Leiden +0.020 [+0.017, +0.024]	CPM +0.081 [+0.076, +0.087]	L-CPM −0.013 [−0.020, −0.007]	Leiden +0.115 [+0.109, +0.120]
two worlds, α=1	0.05	L-CPM −0.046 [−0.051, −0.039]	L-CPM +0.115 [+0.100, +0.128]	L-CPM −0.285 [−0.303, −0.266]	CPM +0.123 [+0.112, +0.134]
		Link +0.072 [+0.066, +0.078]	Link +0.117 [+0.104, +0.130]	Walk. −0.108 [−0.126, −0.090]	Walk. +0.155 [+0.140, +0.170]
two worlds, α=2	0.05	L-CPM +0.001 [−0.007, +0.008]	L-CPM +0.131 [+0.116, +0.144]	L-CPM −0.200 [−0.217, −0.183]	CPM +0.152 [+0.137, +0.167]
		Link +0.071 [+0.064, +0.077]	Link +0.131 [+0.118, +0.145]	Leiden −0.091 [−0.111, −0.072]	Walk. +0.171 [+0.154, +0.187]
LFR, o_n=0		L-CPM −0.143 [−0.156, −0.130]	Info. −0.156 [−0.170, −0.142]	Leiden −0.183 [−0.200, −0.166]	Info. −0.156 [−0.170, −0.142]
		Leiden −0.143 [−0.156, −0.130]	OSLOM −0.155 [−0.168, −0.142]	L-CPM −0.183 [−0.200, −0.165]	OSLOM −0.156 [−0.169, −0.143]
LFR, o_n=0.1		OSLOM −0.231 [−0.242, −0.220]	OSLOM −0.250 [−0.281, −0.223]	OSLOM −0.325 [−0.344, −0.304]	OSLOM −0.272 [−0.300, −0.245]
		L-CPM −0.148 [−0.160, −0.135]	Info. −0.185 [−0.215, −0.155]	Info. −0.243 [−0.262, −0.219]	Info. −0.190 [−0.216, −0.163]
LFR, o_n=0.3		OSLOM −0.402 [−0.416, −0.388]	OSLOM −0.395 [−0.417, −0.368]	OSLOM −0.584 [−0.593, −0.576]	OSLOM −0.420 [−0.440, −0.399]
		SLPA −0.209 [−0.225, −0.192]	Info. −0.196 [−0.212, −0.179]	Leiden −0.397 [−0.407, −0.387]	SLPA −0.212 [−0.243, −0.184]
LFR, $\langle k \rangle$=10		Leiden −0.330 [−0.343, −0.316]	OSLOM −0.338 [−0.357, −0.321]	Leiden −0.365 [−0.387, −0.344]	OSLOM −0.352 [−0.371, −0.334]
		L-CPM −0.329 [−0.342, −0.315]	Info. −0.267 [−0.285, −0.249]	L-CPM −0.361 [−0.383, −0.341]	Info. −0.273 [−0.293, −0.253]
LFR, o_m=4		OSLOM −0.186 [−0.199, −0.173]	OSLOM −0.141 [−0.165, −0.124]	OSLOM −0.250 [−0.258, −0.240]	OSLOM −0.190 [−0.210, −0.175]
		SLPA −0.152 [−0.166, −0.138]	Info. −0.094 [−0.114, −0.077]	SLPA −0.184 [−0.195, −0.173]	SLPA −0.101 [−0.122, −0.084]
LFR, sizes 50–250		OSLOM −0.486 [−0.504, −0.467]	OSLOM −0.473 [−0.496, −0.450]	OSLOM −0.571 [−0.593, −0.549]	OSLOM −0.474 [−0.495, −0.452]
		Leiden −0.448 [−0.467, −0.427]	Leiden −0.454 [−0.477, −0.429]	Leiden −0.547 [−0.572, −0.522]	Leiden −0.425 [−0.448, −0.398]
LFR, C_{LFR}=0.3		Leiden −0.229 [−0.240, −0.215]	SLPA −0.234 [−0.257, −0.213]	Leiden −0.310 [−0.324, −0.297]	Info. −0.255 [−0.274, −0.234]
		L-CPM −0.222 [−0.233, −0.208]	Info. −0.211 [−0.245, −0.177]	L-CPM −0.306 [−0.318, −0.295]	SLPA −0.254 [−0.276, −0.232]
LFR, C_{LFR}=0.5		Leiden −0.076 [−0.082, −0.070]	LPA −0.105 [−0.115, −0.096]	SLPA −0.138 [−0.150, −0.128]	SLPA −0.118 [−0.131, −0.105]
		L-CPM −0.071 [−0.078, −0.065]	SLPA −0.104 [−0.119, −0.092]	OSLOM −0.131 [−0.140, −0.124]	LPA −0.105 [−0.115, −0.096]

TABLE S15. Control families: Ω of HyperCommon (HC), its triangles-only variant and the best other method under the oracle and label-free protocols, with the reference family each control changes (one assumption at a time; Sec. VI), mean over the noise range of each family ($p \in \{0, 0.1, \ldots, 0.8\}$ for the controls, $p \in \{0, \ldots, 0.6\}$ for the swap controls and their references, $p \in [0, 0.8]$ for the other references). The controls are compared with the methods that lead on the main benchmarks (DEMON, ANGEL, SCAN and SCAN-J are left out). Mark on HC as in Table S6.

	oracle			label-free			reference	HC oracle	HC free
Control	HC	tri.	best other	HC	tri.	best other			
hubs, $(50, k_i \in [4, 32]) \times 40$	0.58^{-}	0.53	0.64 (Leiden)	0.56^{+}	0.52	0.53 (Link)	uniform k=8	0.57	0.55
hubs, $(100, k_i \in [8, 64]) \times 20$	0.71^{-}	0.72	0.84 (Leiden)	0.69^{-}	0.70	0.83 (Leiden)	uniform k=16	0.73	0.72
hubs, two worlds	0.76^{-}	0.75	0.85 (L-CPM)	0.75^{-}	0.73	0.83 (L-CPM)	two worlds	0.78	0.76
split overlap, two worlds, o_n=0.05	$0.78^{=}$	0.77	0.78 (L-CPM)	0.78^{+}	0.76	0.71 (Link)	two worlds	0.78	0.76
split overlap, two worlds, o_n=0.10	$0.72^{=}$	0.72	0.73 (L-CPM)	0.72^{+}	0.71	0.62 (CPM)	two worlds, o_n=0.10	0.72	0.71
split overlap, three big, o_n=0.05	0.79^{+}	0.81	0.75 (Link)	0.79^{+}	0.77	0.74 (Link)	three big	0.80	0.78
split overlap, three big, o_n=0.10	0.73^{+}	0.75	0.66 (CPM)	0.72^{+}	0.69	0.62 (Link)	three big, o_n=0.10	0.71	0.67
swaps, uniform k=8, q=0.1	0.68^{-}	0.56	0.81 (Leiden)	$0.65^{=}$	0.56	0.66 (Leiden)	endpoint, q=0.1	0.68	0.65
swaps, uniform k=8, q=0.3	0.63^{-}	0.41	0.85 (Leiden)	0.57^{-}	0.41	0.71 (Info.)	endpoint, q=0.3	0.62	0.57
swaps, three big, q=0.1	0.92^{-}	0.92	0.95 (L-CPM)	0.91^{+}	0.91	0.85 (CPM)	endpoint, q=0.1	0.92	0.91
swaps, three big, q=0.3	0.84^{-}	0.83	0.96 (L-CPM)	0.84^{-}	0.80	0.94 (L-CPM)	endpoint, q=0.3	0.84	0.83
random geometric, $(50, 8) \times 40$	0.50^{-}	0.43	0.57 (Leiden)	0.48^{+}	0.43	0.46 (Leiden)	uniform k=8	0.57	0.55
random geometric, two worlds	0.73^{-}	0.74	0.84 (L-CPM)	0.73^{-}	0.73	0.79 (L-CPM)	two worlds	0.78	0.76

TABLE S16. Null graphs with triadic closure but no planted communities: number of communities and coverage of each method's label-free cover (mean over ten graphs). RGG: random geometric graph on the unit torus, 2000 nodes, mean degree 8 (clustering 0.59); HK: Holme–Kim graph, 2000 nodes, $m = 4$, triangle probability 0.9 (clustering 0.43).

Graph		HC	tri.	Link	CPM	Leiden	L-CPM	Info.	Walk.	LPA	SBM	SLPA	OSLOM	DEMON	ANGEL	SCAN	SCAN-J
RGG	communities	101	98	188	162	34	35	36	53	178	32	172	36	237	89	92	66
	coverage	0.98	0.98	0.99	0.93	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.98	0.96	0.97
HK	communities	175	166	391	288	16	16	28	32	55	12	42	13	196	156	97	121
	coverage	0.96	0.85	1.00	0.66	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.76	0.82	0.82

TABLE S17. HyperCommon with each commonality function, best threshold per graph: mean Ω over the noise range ($p \in [0, 0.8]$ for rings, $\mu \in [0.1, 0.6]$ for LFR), ten test graphs per noise level, the threshold a quantile of each function's triple scores. Superscripts mark a paired 95% interval of the difference from Jaccard that excludes zero ($^+$ better, $^-$ worse). Bold: best single function in the row (last column excluded). $I/\max$, $I/\min$: overlap normalised by the larger or smaller neighbourhood; I: overlap count; RA: resource allocation. Last column: function and level chosen together on each graph by the same rule. Quantile levels $0, 0.04, \ldots, 0.96$ and $0.97, 0.98, 0.99, 0.995$; rings sampled every 0.1 in p. Jaccard values can differ from the HC columns of Tables S6 and S8 (by up to 0.04) because thresholds are quantile levels and rings are sampled every 0.1 in p.

Family	Jaccard	cosine	$I/\max$	$I/\min$	I^2/U	I	RA	per graph
uniform, k=8, o_n=0	**0.62**	**0.62**	0.61$^-$	0.61$^-$	**0.62**	0.56$^-$	0.57$^-$	0.63
uniform, k=8, o_n=0.05	**0.58**	**0.58**	0.56$^-$	0.54$^-$	0.57$^-$	0.52$^-$	0.49$^-$	0.59
uniform, k=8, o_n=0.10	0.51	0.51	0.50$^-$	**0.52**	0.50$^-$	0.46$^-$	0.43$^-$	0.53
uniform, k=16, o_n=0	0.78	**0.79**$^+$	0.77$^-$	0.77$^-$	0.78	0.74$^-$	0.74$^-$	0.79
uniform, k=16, o_n=0.05	**0.72**	**0.72**	**0.72**	0.71$^-$	0.70$^-$	0.63$^-$	0.62$^-$	0.74
uniform, k=16, o_n=0.10	**0.64**	0.63$^-$	**0.64**	0.62$^-$	0.58$^-$	0.43$^-$	0.45$^-$	0.67
equal k, unequal s (k=8)	**0.52**	**0.52**	0.51$^-$	0.50$^-$	**0.52**	0.46$^-$	0.44$^-$	0.53
equal k, unequal s (k=16)	**0.74**	**0.74**	**0.74**	0.73	0.73$^-$	0.67$^-$	0.66$^-$	0.76
equal s, unequal k	0.73	0.73$^-$	**0.74**	0.64$^-$	0.65$^-$	0.52$^-$	0.57$^-$	0.75
two worlds	0.80	0.78$^-$	**0.81**	0.66$^-$	0.75$^-$	0.67$^-$	0.66$^-$	0.82
three big, many small	0.81	0.80$^-$	**0.83**$^+$	0.68$^-$	0.77$^-$	0.62$^-$	0.63$^-$	0.85
gradient of ten rings	0.59	0.58$^-$	**0.60**$^+$	0.51$^-$	0.58	0.52$^-$	0.52$^-$	0.65
LFR, o_n=0	0.84	**0.85**$^+$	0.81$^-$	0.84	**0.85**	0.76$^-$	0.72$^-$	0.87
LFR, o_n=0.1	0.73	**0.74**$^+$	0.71$^-$	**0.74**$^+$	0.70$^-$	0.59$^-$	0.53$^-$	0.77
LFR, o_n=0.3	0.49	**0.50**	0.48$^-$	**0.50**	0.43$^-$	0.32$^-$	0.30$^-$	0.53
LFR, $\langle k \rangle$=10	0.47	**0.48**$^+$	0.42$^-$	0.43$^-$	0.47	0.40$^-$	0.26$^-$	0.49
LFR, o_m=4	0.61	0.61	0.60$^-$	**0.62**$^+$	0.58$^-$	0.47$^-$	0.45$^-$	0.63
LFR, sizes 50–250	0.45	**0.46**$^+$	0.41$^-$	**0.46**	**0.46**	0.29$^-$	0.25$^-$	0.51
better / worse than Jaccard	–	5 / 5	2 / 11	2 / 11	0 / 11	0 / 18	0 / 18	

TABLE S18. HyperCommon with each commonality function, label-free threshold (largest overlapping modularity): mean Ω over the noise range ($p \in [0, 0.8]$ for rings, $\mu \in [0.1, 0.6]$ for LFR), ten test graphs per noise level, the threshold a quantile of each function's triple scores. Superscripts mark a paired 95% interval of the difference from Jaccard that excludes zero ($^+$ better, $^-$ worse). Bold: best single function in the row (last column excluded). $I/\max$, $I/\min$: overlap normalised by the larger or smaller neighbourhood; I: overlap count; RA: resource allocation. Last column: function and level chosen together on each graph by the same rule. Quantile levels $0, 0.04, \ldots, 0.96$ and $0.97, 0.98, 0.99, 0.995$; rings sampled every 0.1 in p. Jaccard values can differ from the HC columns of Tables S6 and S8 (by up to 0.04) because thresholds are quantile levels and rings are sampled every 0.1 in p.

Family	Jaccard	cosine	$I/\max$	$I/\min$	I^2/U	I	RA	per graph
uniform, k=8, o_n=0	0.59	0.60$^+$	0.59	**0.61**$^+$	0.58	0.55$^-$	0.54$^-$	0.61
uniform, k=8, o_n=0.05	**0.56**	**0.56**	0.55	0.50$^-$	0.55$^-$	0.51$^-$	0.45$^-$	0.56
uniform, k=8, o_n=0.10	**0.48**	**0.48**	**0.48**	0.47	0.46$^-$	0.44$^-$	0.40$^-$	0.48
uniform, k=16, o_n=0	0.78	**0.79**$^+$	0.77$^-$	0.77$^-$	0.78	0.74$^-$	0.74$^-$	0.79
uniform, k=16, o_n=0.05	**0.71**	**0.71**	**0.71**	0.69$^-$	0.69$^-$	0.62$^-$	0.60$^-$	0.72
uniform, k=16, o_n=0.10	**0.63**	0.62$^-$	0.62	0.60$^-$	0.56$^-$	0.41$^-$	0.42$^-$	0.65
equal k, unequal s (k=8)	0.51	**0.52**	0.50$^-$	0.49$^-$	0.51	0.45$^-$	0.43$^-$	0.52
equal k, unequal s (k=16)	0.72	**0.73**	**0.73**	0.71$^-$	0.70$^-$	0.64$^-$	0.64$^-$	0.73
equal s, unequal k	0.71	0.70$^-$	**0.72**$^+$	0.62$^-$	0.62$^-$	0.48$^-$	0.55$^-$	0.73
two worlds	0.78	0.77$^-$	**0.80**	0.63$^-$	0.71$^-$	0.66$^-$	0.63$^-$	0.81
three big, many small	0.78	0.77	**0.82**$^+$	0.62$^-$	0.63$^-$	0.44$^-$	0.49$^-$	0.81
gradient of ten rings	0.36	0.36	0.39$^+$	0.32	**0.50**$^+$	0.48$^+$	0.42	0.31
LFR, o_n=0	0.81	0.82	0.80	**0.84**$^+$	0.81	0.70$^-$	0.67$^-$	0.82
LFR, o_n=0.1	0.70	**0.72**$^+$	0.67$^-$	0.69	0.65$^-$	0.56$^-$	0.45$^-$	0.73
LFR, o_n=0.3	0.43	0.43	**0.44**	**0.44**	0.37$^-$	0.26$^-$	0.28$^-$	0.45
LFR, $\langle k \rangle$=10	0.41	**0.44**	0.38$^-$	0.37$^-$	0.41	0.40	0.21$^-$	0.43
LFR, o_m=4	0.60	0.60	0.59	**0.61**	0.57$^-$	0.45$^-$	0.41$^-$	0.62
LFR, sizes 50–250	0.43	**0.44**	0.40$^-$	**0.44**	**0.44**	0.28$^-$	0.25$^-$	0.48
better / worse than Jaccard	–	3 / 3	3 / 5	2 / 10	1 / 11	1 / 16	0 / 17	

S6. REAL NETWORKS

TABLE S19. Wedges of thirty-two SNAP networks, read as simple undirected graphs: nodes n, edges m, largest degree $d_{\max}$, and the number of wedges $W = \sum_v \binom{d_v}{2}$ against $m \ln m$, without and with the degree-ratio pruning of Sec. V C at $t_{\min} = 0.02$, which is exact for every threshold $t \geq 0.02$. The last three are the graphs of Table III; the others have no ground truth.

Network	Kind	n	m	$d_{\max}$	$W/(m \ln m)$	Pruned
ca-GrQc	collaboration	5 241	14 484	81	1.66	1.66
ca-HepTh	collaboration	9 875	25 973	65	1.13	1.13
ca-HepPh	collaboration	12 006	118 489	491	11.04	10.91
ca-AstroPh	collaboration	18 771	198 050	504	5.28	5.26
ca-CondMat	collaboration	23 133	93 439	279	1.84	1.83
email-Enron	communication	36 692	183 831	1 383	11.47	6.39
email-EuAll	communication	265 009	364 481	7 636	41.84	1.43
wiki-Talk	communication	2 394 385	4 659 565	100 029	176.02	3.69
ego-Facebook	social	4 039	88 234	1 045	9.27	8.68
ego-Twitter	social	81 306	1 342 296	3 383	12.15	9.34
wiki-Vote	social	7 115	100 762	1 065	12.53	11.94
soc-Epinions1	social	75 879	405 740	3 044	14.16	10.19
soc-Slashdot	social	82 168	504 230	2 552	11.33	7.69
loc-Brightkite	social	58 228	214 078	1 134	5.11	3.95
loc-Gowalla	social	196 591	950 327	14 730	22.20	4.81
com-LiveJournal	social	3 997 962	34 681 189	14 815	7.07	5.79
as-skitter	internet	1 696 415	11 095 298	35 455	89.02	4.66
as-CAIDA	internet	26 475	53 381	2 628	25.65	1.95
oregon1-010526	internet	11 174	23 409	2 389	26.30	2.04
p2p-Gnutella31	internet	62 586	147 892	95	0.89	0.89
web-Google	web	875 713	4 322 051	6 332	11.02	2.13
web-Stanford	web	281 903	1 992 636	38 625	136.46	4.91
web-NotreDame	web	325 729	1 090 108	10 721	20.12	3.27
web-BerkStan	web	685 230	6 649 470	84 230	267.87	9.08
roadNet-CA	road	1 965 206	2 766 607	12	0.15	0.15
roadNet-PA	road	1 088 092	1 541 898	9	0.15	0.15
cit-HepPh	citation	34 546	420 877	846	4.83	4.72
cit-Patents	citation	3 774 768	16 518 947	793	1.22	1.22
amazon0601	co-purchase	403 394	2 443 408	2 752	2.01	1.58
com-amazon	co-purchase	334 863	925 872	549	0.77	0.74
com-dblp	collaboration	317 080	1 049 866	343	1.50	1.48
com-youtube	social	1 134 890	2 987 624	28 754	33.10	3.70

[1] A. Lancichinetti and S. Fortunato, Benchmarks for testing community detection algorithms on directed and weighted graphs with overlapping communities, Physical Review E **80**, 016118 (2009).

[2] G. Rossetti, L. Milli, and R. Cazabet, CDLIB: a python library to extract, compare and evaluate communities from complex networks, Applied Network Science **4**, 52 (2019).

[3] V. A. Traag, leidenalg: Implementation of the Leiden algorithm in C++ for Python, version 0.11.0, https://github.com/vtraag/leidenalg.

[4] D. Edler, A. Holmgren, and M. Rosvall, The MapEquation software package, infomap version 2.10.1, https://mapequation.org.

[5] T. P. Peixoto, The graph-tool python library, figshare (2014).

TABLE S20. Detection on fourteen further networks with recorded groups (Netzschleuder repository [21]; sizes in Table S2), scored by best-match F1 on the labelled nodes, to which detected communities are restricted; each method at its best parameter (best) and at the parameter of largest overlapping modularity (free); stochastic methods: mean over five seeds. HC: HyperCommon with Jaccard; HC$_f$: the seven commonality functions together, each on 100 quantile levels of its own triple scores (in brackets the function chosen; J when Jaccard ties), the free choice taken over all of them; with labels it chooses among 700 (function, threshold) pairs per network, so its best-parameter column is an upper bound for the family, not one method's parameter grid (with 30 quantile levels per function its four wins are unchanged). "Best other": the most accurate of the thirteen other methods under that protocol. Bold: best of HC, HC$_f$ and the other methods under each protocol. Functions: J, Jaccard; cos, cosine; $I/\min$ and $I/\max$, overlap normalised by the smaller or larger neighbourhood; I, overlap count; RA, resource allocation. College football: the conference labels of Ref. [22] are those of the 2001 season while the games are from the 2000 season; corrected labels are given in Ref. [23]. Every method on every network is in Tables S22 and S23.

	best parameter			free (label-free)		
Network	HC	HC$_f$	best other	HC	HC$_f$	best other
College football	0.909	**0.914** (RA)	0.896 (OSLOM)	0.875	0.886 ($I/\min$)	**0.896** (OSLOM)
Political books	0.784	0.786 ($I/\min$)	**0.807** (OSLOM)	0.705	0.699 (cos)	**0.767** (SCAN)
Political blogs	0.636	0.636 (J)	**0.642** (OSLOM)	0.195	0.211 (cos)	**0.642** (OSLOM)
Primary school	0.405	0.826 ($I/\min$)	**0.893** (Leiden)	0.405	0.826 ($I/\min$)	**0.893** (L-CPM)
Facebook ego 0	0.399	**0.442** (I^2/U)	0.419 (SCAN)	0.390	0.378 (cos)	**0.410** (SCAN-J)
Facebook ego 107	0.552	0.616 (RA)	**0.708** (SLPA)	0.552	0.522 (cos)	**0.697** (SLPA)
Facebook ego 348	0.651	0.661 ($I/\min$)	**0.673** (OSLOM)	0.479	0.479 (J)	**0.647** (OSLOM)
Facebook ego 414	0.733	0.758 ($I/\min$)	**0.760** (SCAN)	0.680	**0.757** ($I/\min$)	**0.757** (L-CPM)
Facebook ego 686	0.578	**0.674** (I^2/U)	0.633 (ANGEL)	0.556	0.556 (J)	**0.620** (DEMON)
Facebook ego 698	0.826	**0.866** (I^2/U)	0.849 (CPM)	0.826	0.831 ($I/\min$)	**0.849** (CPM)
Facebook ego 1684	0.612	0.624 ($I/\min$)	**0.633** (Infomap)	0.605	0.619 (cos)	**0.633** (Infomap)
Facebook ego 1912	0.463	0.484 (cos)	**0.532** (Leiden)	0.369	0.400 ($I/\max$)	**0.525** (Leiden)
Facebook ego 3437	0.564	0.586 (cos)	**0.698** (Leiden)	0.526	0.551 (cos)	**0.644** (Leiden)
Facebook ego 3980	0.681	0.692 (cos)	**0.734** (LPA)	0.567	0.605 (cos)	**0.734** (LPA)

TABLE S21. As Table S20, scored by Ω on the labelled nodes.

	best parameter			free (label-free)		
Network	HC	HC$_f$	best other	HC	HC$_f$	best other
College football	0.919	**0.927** (RA)	0.906 (Infomap)	0.836	**0.889** ($I/\min$)	0.868 (OSLOM)
Political books	0.687	**0.709** (cos)	0.704 (OSLOM)	0.659	**0.680** (cos)	0.676 (SLPA)
Political blogs	0.041	0.053 (cos)	**0.552** (Leiden)	0.041	0.053 (cos)	**0.543** (OSLOM)
Primary school	0.060	0.713 ($I/\min$)	**0.812** (Leiden)	0.060	0.713 ($I/\min$)	**0.812** (L-CPM)
Facebook ego 0	0.286	0.339 ($I/\min$)	**0.349** (OSLOM)	0.251	0.258 (cos)	**0.338** (Infomap)
Facebook ego 107	0.513	0.542 (cos)	**0.844** (L-CPM)	0.513	0.526 (cos)	**0.736** (LPA)
Facebook ego 348	0.161	0.184 (I)	**0.291** (DEMON)	−0.082	−0.082 (J)	**0.142** (DEMON)
Facebook ego 414	0.861	0.861 (J)	**0.872** (L-CPM)	**0.861**	0.810 ($I/\min$)	0.853 (SCAN)
Facebook ego 686	0.107	**0.163** (RA)	0.110 (Link)	0.075	0.075 (J)	**0.093** (SCAN-J)
Facebook ego 698	0.588	**0.712** ($I/\min$)	0.687 (Link)	0.588	0.621 ($I/\min$)	**0.647** (Leiden)
Facebook ego 1684	0.744	0.785 ($I/\min$)	**0.815** (Infomap)	0.744	0.767 (cos)	**0.815** (Infomap)
Facebook ego 1912	0.576	0.590 ($I/\max$)	**0.671** (OSLOM)	0.574	0.590 ($I/\max$)	**0.666** (OSLOM)
Facebook ego 3437	0.462	0.486 (I^2/U)	**0.504** (Walktrap)	0.462	0.450 (cos)	**0.504** (Walktrap)
Facebook ego 3980	0.190	0.190 (J)	**0.535** (LPA)	0.043	0.130 (cos)	**0.535** (LPA)

[6] V. A. Traag, P. Van Dooren, and Y. Nesterov, Narrow scope for resolution-limit-free community detection, Physical Review E **84**, 016114 (2011).

[7] Y.-Y. Ahn, J. P. Bagrow, and S. Lehmann, Link communities reveal multiscale complexity in networks, Nature **466**, 761 (2010).

[8] J. M. Kumpula, M. Kivelä, K. Kaski, and J. Saramäki, Sequential algorithm for fast clique percolation, Physical Review E **78**, 026109 (2008).

[9] A. Arasu, V. Ganti, and R. Kaushik, Efficient exact set-similarity joins, in *Proceedings of the 32nd International Conference on Very Large Data Bases* (2006) pp. 918–929.

[10] R. J. Bayardo, Y. Ma, and R. Srikant, Scaling up all pairs similarity search, in *Proceedings of the 16th International*

TABLE S22. Every method on the fourteen smaller networks: best-match F1 at the best parameter (stochastic methods: mean over five seeds), with each method's rank among the methods shown (ablations included) in the last row as the mean rank over the fourteen networks. Bold: best in row, within 0.005.

Network	HC	e-conn.	tri.	2-test	Link	CPM	Leiden	L-CPM	Infomap	LPA	Walktrap	SLPA	DEMON	ANGEL	SCAN	SCAN-J	OSLOM
College football	0.91	0.91	0.91	**0.91**	0.68	0.85	0.89	0.89	0.89	0.83	0.87	0.77	0.81	0.75	0.89	0.89	0.90
Political books	0.78	0.74	0.66	0.78	0.71	0.65	0.79	0.79	0.78	0.62	0.65	0.71	0.75	0.75	0.77	0.66	**0.81**
Political blogs	0.64	0.64	0.35	0.64	0.64	0.35	0.43	0.37	0.50	0.43	0.42	0.57	0.60	0.59	0.62	0.54	**0.64**
Primary school	0.40	0.40	0.44	0.42	0.52	0.19	**0.89**	**0.89**	0.19	0.19	0.55	0.19	0.68	0.64	0.51	0.46	**0.89**
Facebook ego 0	0.40	0.40	0.40	**0.41**	0.38	0.40	0.38	0.36	0.40	0.36	0.34	0.39	**0.42**	0.39	**0.42**	0.41	0.37
Facebook ego 107	0.55	0.53	0.52	0.60	0.48	0.64	0.62	0.59	0.69	0.64	0.62	**0.71**	0.59	0.55	0.63	0.62	0.54
Facebook ego 348	0.65	0.65	0.63	0.66	0.61	0.66	0.66	0.66	0.66	0.51	0.46	**0.67**	0.61	0.66	0.66	0.65	**0.67**
Facebook ego 414	0.73	0.73	**0.76**	0.75	0.62	0.75	**0.76**	**0.76**	**0.76**	**0.76**	0.67	**0.76**	0.70	0.73	**0.76**	0.75	**0.76**
Facebook ego 686	0.58	0.58	**0.68**	0.62	0.55	0.55	0.61	0.55	0.62	0.55	0.42	0.55	0.62	0.63	0.62	0.61	0.59
Facebook ego 698	0.83	0.83	0.83	0.82	0.76	**0.85**	0.81	0.81	0.82	0.81	0.77	0.80	0.82	0.76	0.82	0.83	0.82
Facebook ego 1684	0.61	0.61	0.59	**0.63**	0.42	0.48	0.62	0.56	**0.63**	0.57	0.55	0.62	0.54	0.57	**0.63**	0.62	0.55
Facebook ego 1912	0.46	0.46	0.41	0.45	0.42	–	**0.53**	0.45	0.49	0.47	0.47	0.52	0.49	0.48	0.44	0.44	0.47
Facebook ego 3437	0.56	0.56	0.56	0.61	0.52	0.51	**0.70**	**0.70**	0.68	0.54	0.56	0.54	0.62	0.59	0.61	0.61	0.62
Facebook ego 3980	0.68	0.68	0.68	0.68	0.68	0.61	0.71	0.68	0.68	**0.73**	0.62	0.66	0.63	0.63	0.68	0.68	0.68
mean rank	8.1	8.6	9.2	6.6	13.4	11.5	6.0	8.8	6.2	11.5	13.1	9.3	8.9	10.0	6.2	8.3	6.9

TABLE S23. Every method on the fourteen smaller networks: best-match F1 at the label-free parameter (stochastic methods: mean over five seeds), with each method's rank among the methods shown (ablations included) in the last row as the mean rank over the fourteen networks. Bold: best in row, within 0.005.

Network	HC	e-conn.	tri.	2-test	Link	CPM	Leiden	L-CPM	Infomap	LPA	Walktrap	SLPA	DEMON	ANGEL	SCAN	SCAN-J	OSLOM
College football	0.88	0.88	0.86	**0.89**	0.42	0.85	0.87	0.83	0.87	0.83	0.87	0.77	0.69	0.74	**0.89**	**0.89**	**0.90**
Political books	0.71	0.70	0.59	**0.78**	0.71	0.65	0.66	0.72	0.62	0.62	0.65	0.69	0.71	0.62	0.77	0.66	0.69
Political blogs	0.20	0.19	0.18	0.18	0.29	0.23	0.42	0.37	0.47	0.43	0.42	0.57	0.43	0.49	0.19	0.12	**0.64**
Primary school	0.40	0.40	0.44	0.40	0.31	0.19	0.77	**0.89**	0.19	0.19	0.55	0.19	0.35	0.64	0.48	0.46	0.88
Facebook ego 0	0.39	0.38	0.38	0.39	0.35	0.38	0.37	0.35	0.39	0.36	0.34	0.38	0.40	0.35	0.40	**0.41**	0.36
Facebook ego 107	0.55	0.53	0.52	0.52	0.35	0.64	0.60	0.55	0.63	0.64	0.62	**0.70**	0.59	0.45	0.44	0.48	0.51
Facebook ego 348	0.48	0.48	0.48	0.53	0.50	0.47	0.53	0.46	0.56	0.51	0.46	**0.64**	0.58	0.56	0.51	0.51	**0.65**
Facebook ego 414	0.68	0.68	**0.75**	0.70	0.58	0.75	0.67	**0.76**	0.68	**0.76**	0.67	**0.76**	0.70	0.73	0.70	0.70	**0.76**
Facebook ego 686	0.56	0.56	0.58	0.61	0.42	0.46	0.54	0.37	0.30	0.55	0.42	0.54	**0.62**	0.50	**0.62**	0.61	0.56
Facebook ego 698	0.83	0.83	0.83	0.82	0.76	**0.85**	0.73	0.81	0.72	0.81	0.77	0.80	0.82	0.73	0.81	0.82	0.81
Facebook ego 1684	0.60	0.59	0.58	0.60	0.26	0.45	0.61	0.50	**0.63**	0.57	0.55	0.61	0.52	0.57	0.62	0.62	0.49
Facebook ego 1912	0.37	0.37	0.36	0.42	0.32	–	**0.53**	0.40	0.49	0.47	0.47	0.51	0.44	0.36	0.38	0.38	0.45
Facebook ego 3437	0.53	0.53	0.53	0.53	0.45	0.51	**0.64**	0.56	0.55	0.54	0.56	0.54	0.56	0.52	0.57	0.55	0.58
Facebook ego 3980	0.57	0.57	0.57	0.50	0.60	0.61	0.68	0.53	0.61	**0.73**	0.62	0.66	0.57	0.50	0.51	0.50	0.65
mean rank	8.9	9.6	10.2	8.4	13.6	10.3	7.4	9.5	8.8	8.6	10.0	6.6	7.1	11.6	7.4	8.4	6.2

Conference on World Wide Web (2007) pp. 131–140.
[11] D. Hric, R. K. Darst, and S. Fortunato, Community detection in networks: Structural communities versus ground truth, Physical Review E **90**, 062805 (2014).
[12] J. Yang and J. Leskovec, Defining and evaluating network communities based on ground-truth, Knowledge and Information Systems **42**, 181 (2015).
[13] P. Jaccard, The distribution of the flora in the alpine zone, New Phytologist **11**, 37 (1912).
[14] D. Deinega, *Overlapping Communities Detection in Complex Networks*, Master's thesis, Ariel University (2023).
[15] G. Salton and M. J. McGill, *Introduction to Modern Information Retrieval* (McGraw-Hill, 1983).
[16] X. Xu, N. Yuruk, Z. Feng, and T. A. J. Schweiger, SCAN: A structural clustering algorithm for networks, in *Proceedings of the 13th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (2007) pp. 824–833.
[17] T. Zhou, L. Lü, and Y.-C. Zhang, Predicting missing links via local information, The European Physical Journal B **71**, 623 (2009).
[18] L. Lü and T. Zhou, Link prediction in complex networks: A survey, Physica A: Statistical Mechanics and its Applications **390**, 1150 (2011).
[19] E. Ravasz, A. L. Somera, D. A. Mongru, Z. N. Oltvai, and A.-L. Barabási, Hierarchical organization of modularity in metabolic networks, Science **297**, 1551 (2002).
[20] G. Palla, I. Derényi, I. Farkas, and T. Vicsek, Uncovering the overlapping community structure of complex networks in nature and society, Nature **435**, 814 (2005).
[21] T. P. Peixoto, The Netzschleuder network catalogue and repository (2020), https://networks.skewed.de.
[22] M. Girvan and M. E. J. Newman, Community structure in social and biological networks, Proceedings of the National Academy of Sciences **99**, 7821 (2002).
[23] T. S. Evans, Clique graphs and overlapping communities, Journal of Statistical Mechanics: Theory and Experiment **2010**, P12037 (2010).